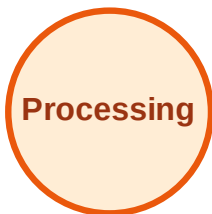


BFS Traversal

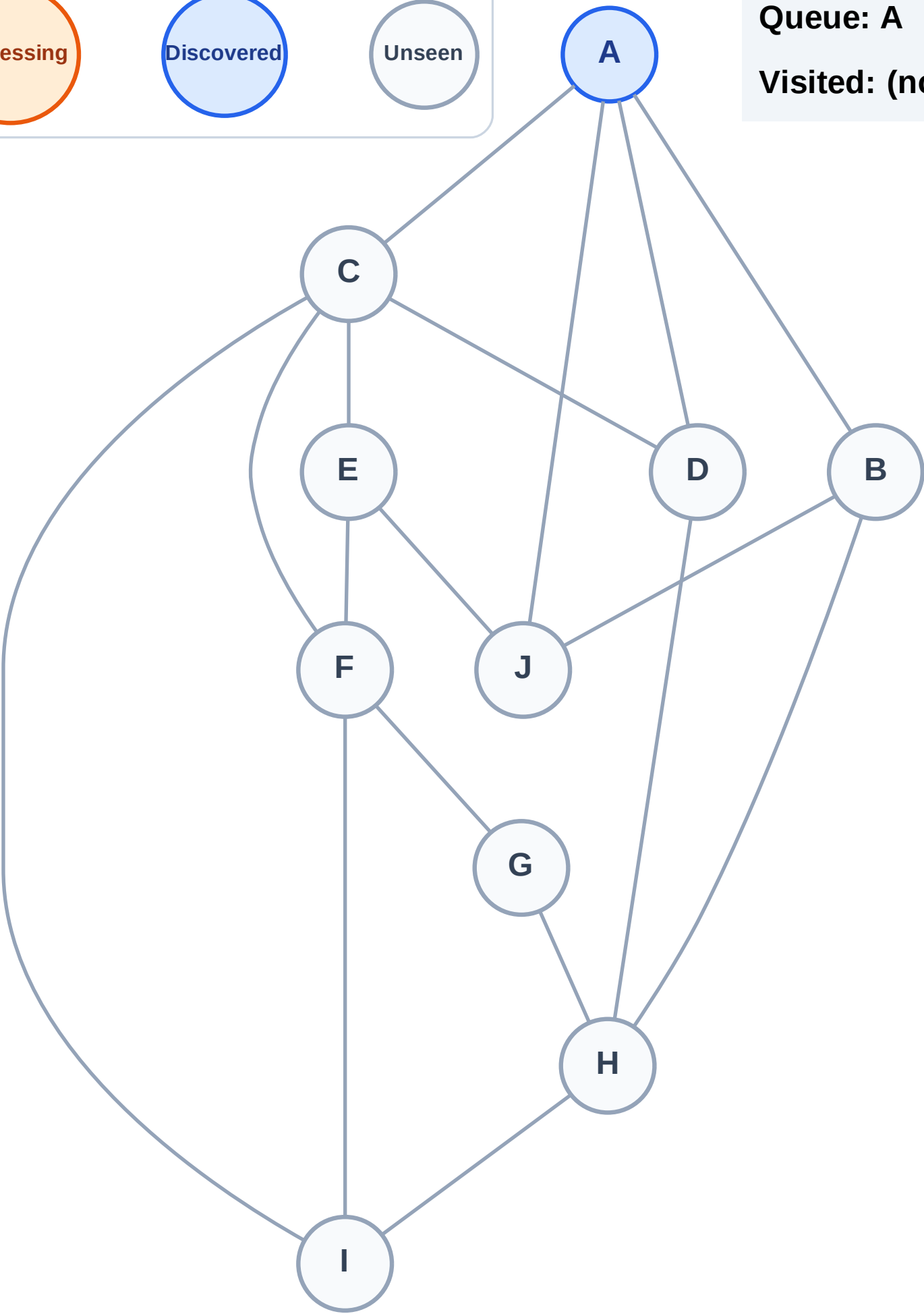
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

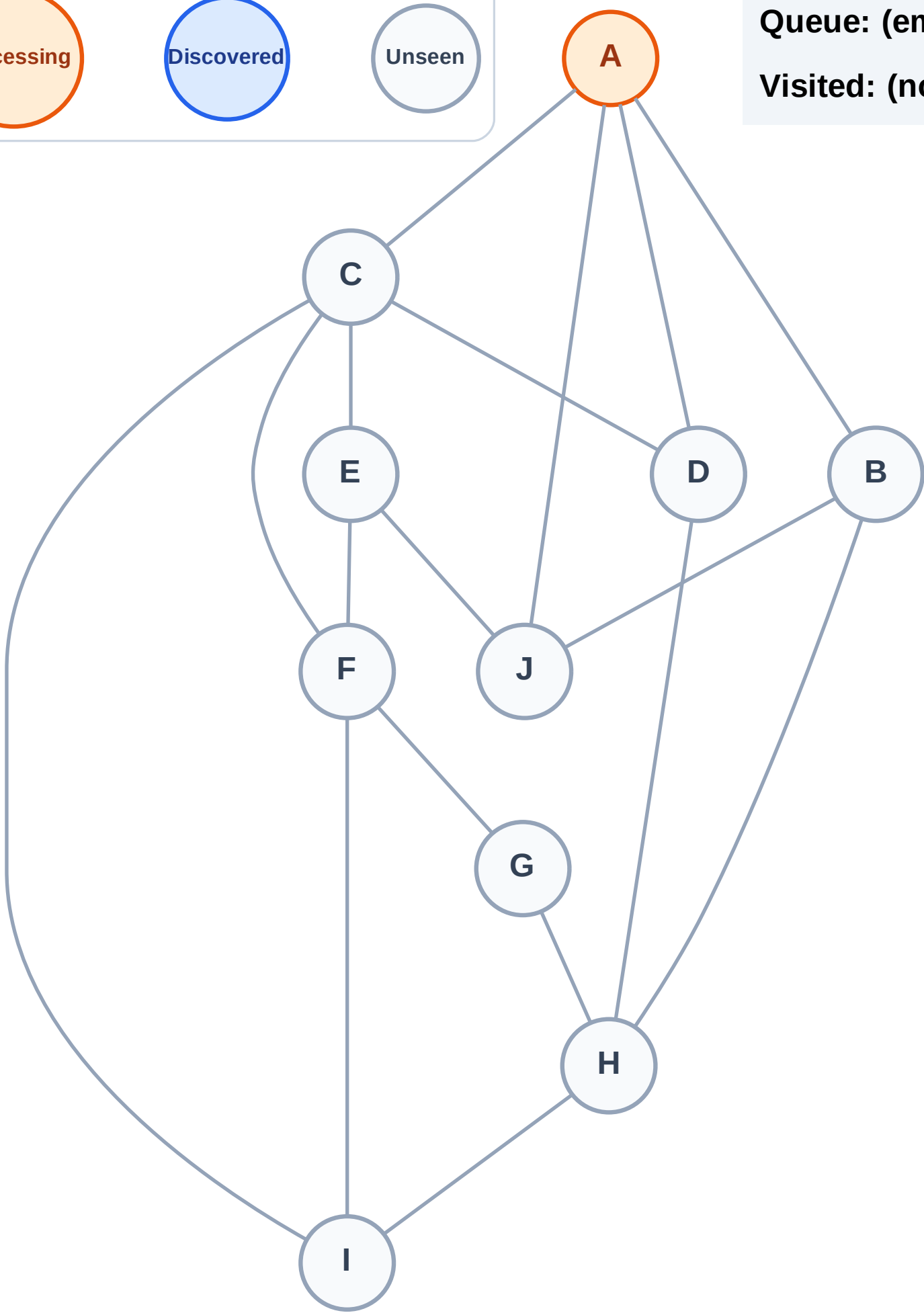
Step 2: Process node A

Legend



Queue: (empty)

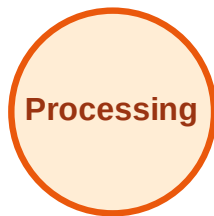
Visited: (none)



BFS Traversal

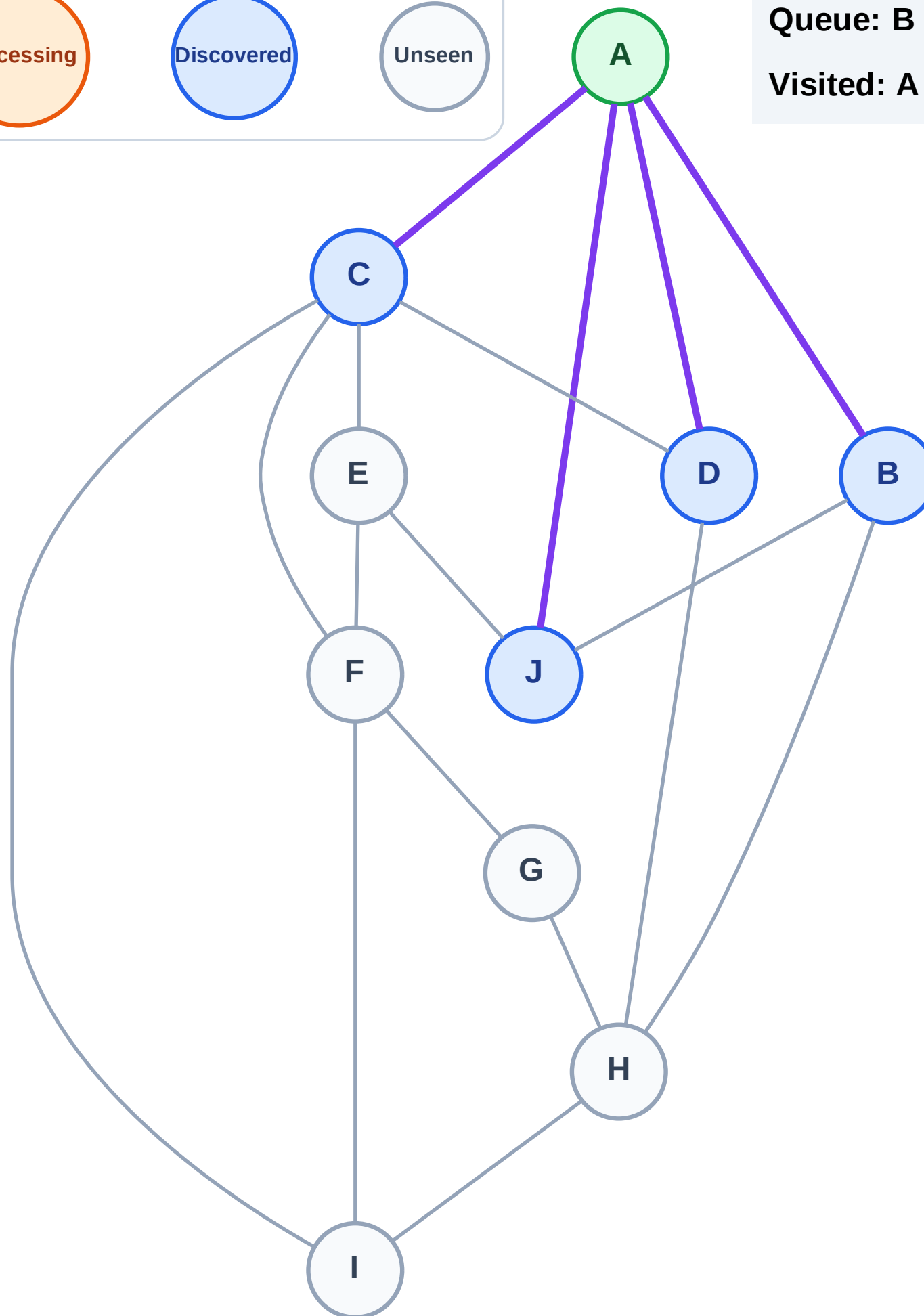
Step 3: Discover B, C, D, J from A

Legend



Queue: B → C → D → J

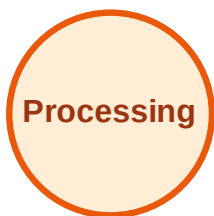
Visited: A



BFS Traversal

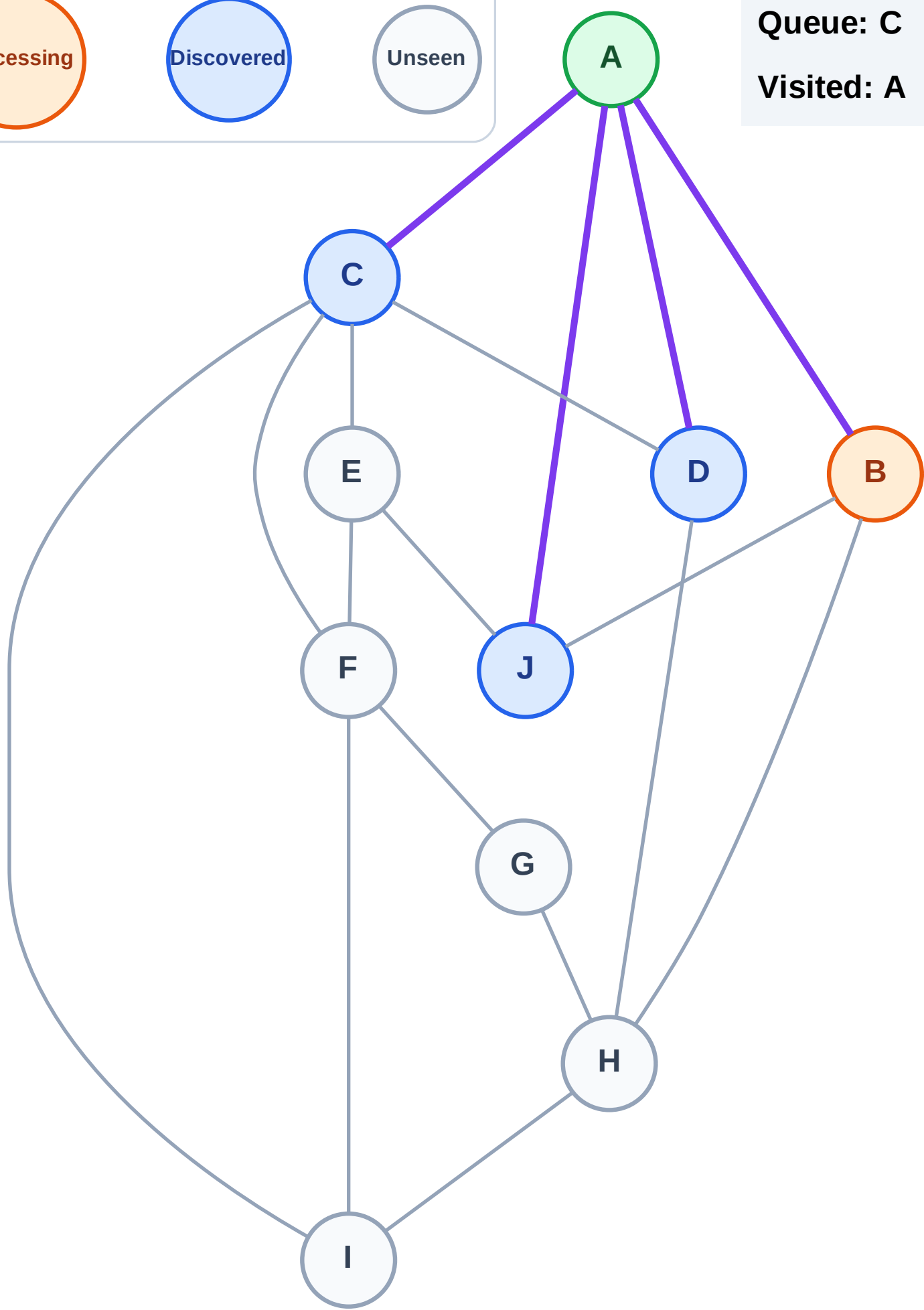
Step 4: Process node B

Legend



Queue: C → D → J

Visited: A



BFS Traversal

Step 5: Discover H from B

Legend

Complete

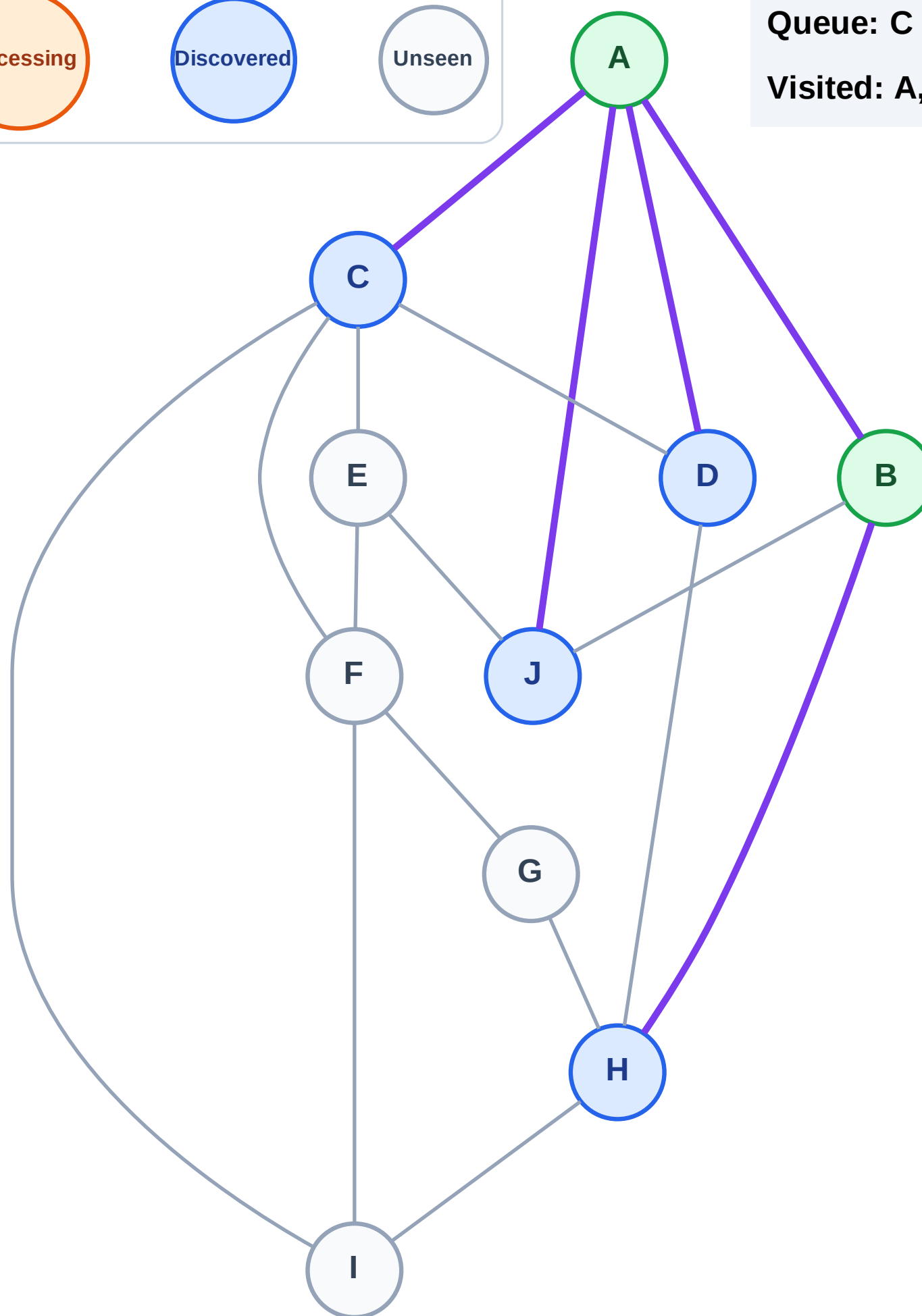
Processing

Discovered

Unseen

Queue: C → D → J → H

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

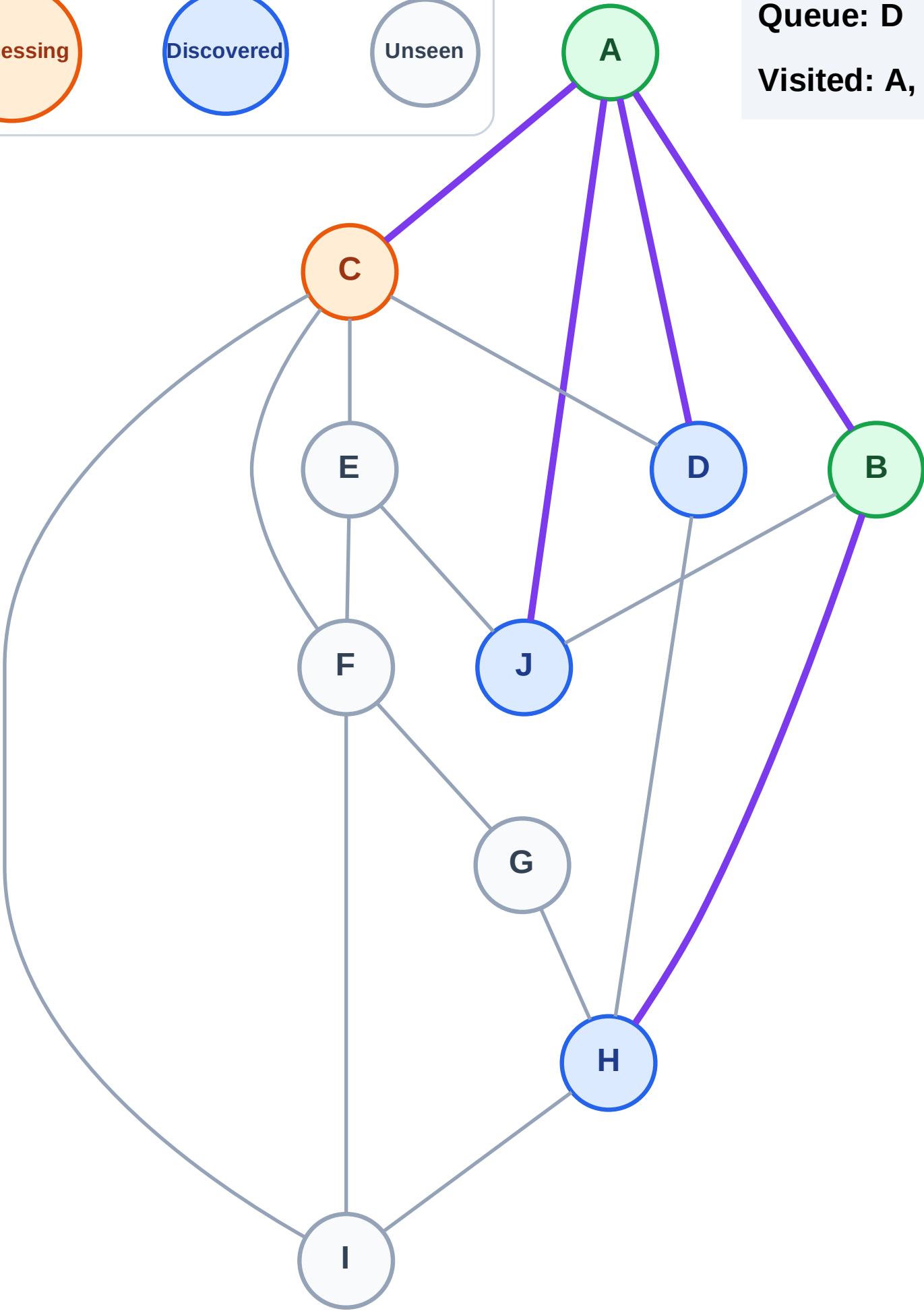
Processing

Discovered

Unseen

Queue: D → J → H

Visited: A, B



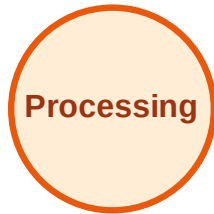
BFS Traversal

Step 7: Discover E, F, I from C

Legend



Complete



Processing

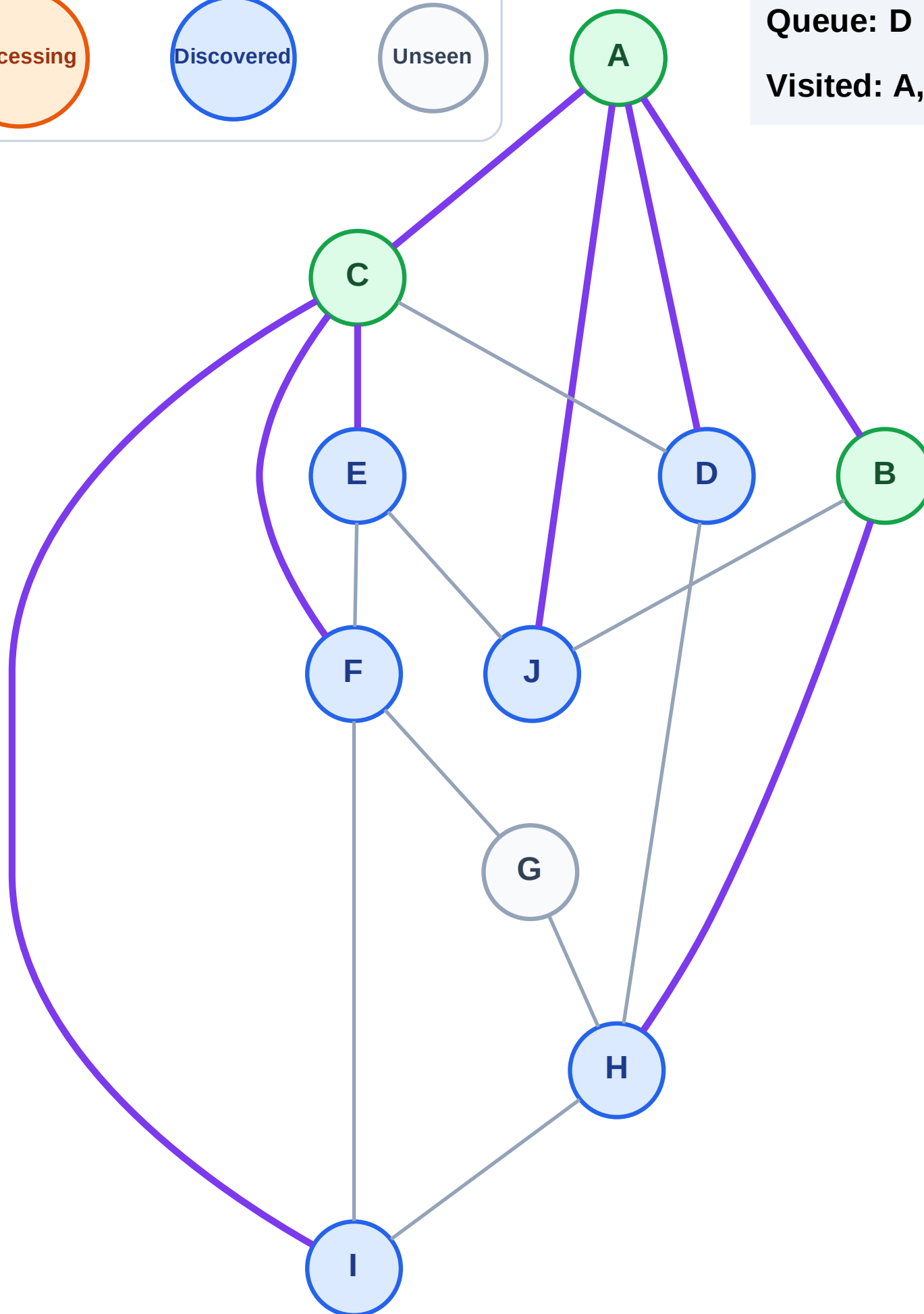


Discovered



Unseen

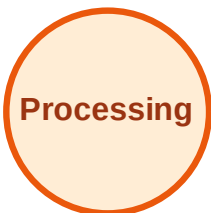
Queue: D → J → H → E → F → I
Visited: A, B, C



BFS Traversal

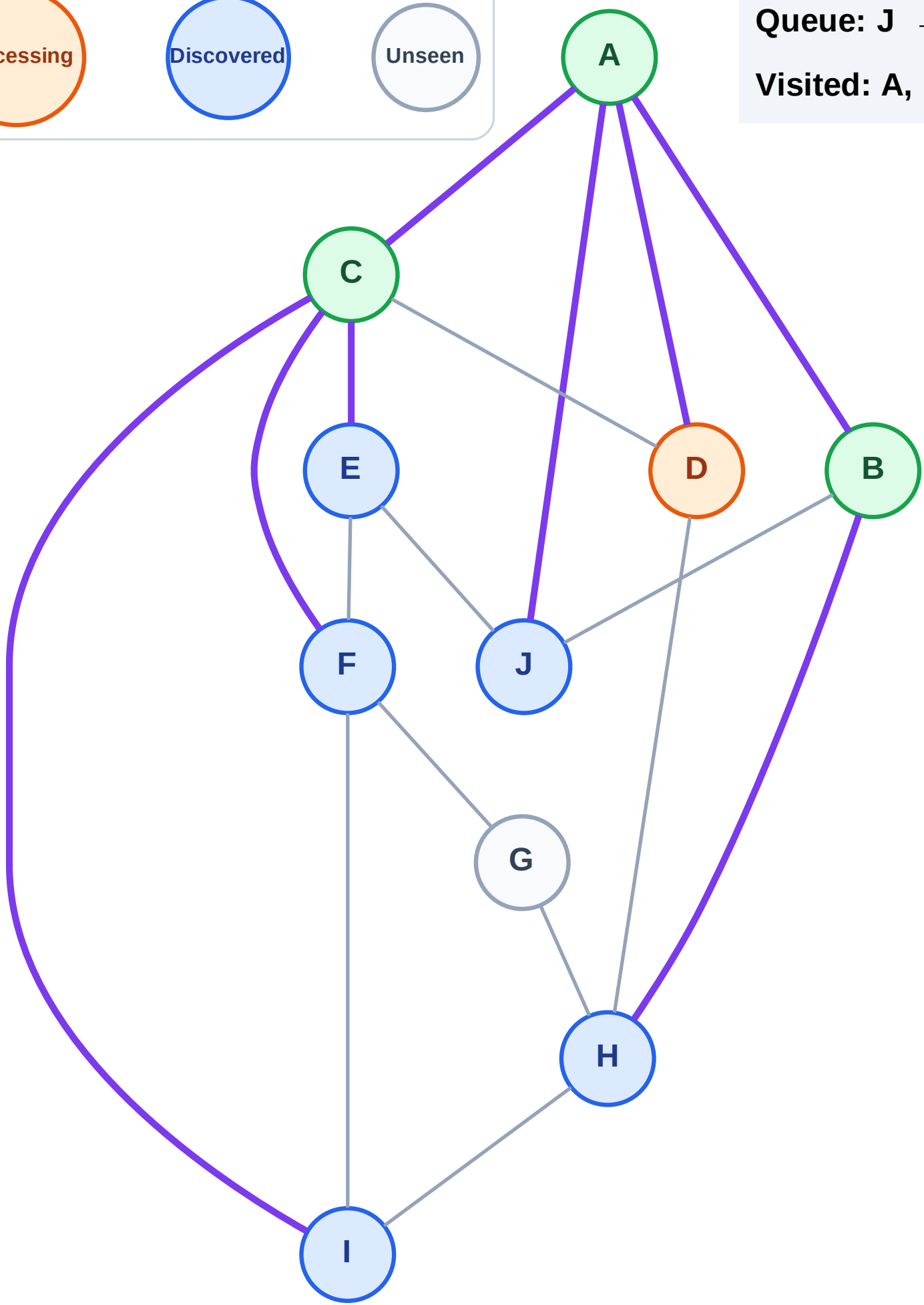
Step 8: Process node D

Legend



Queue: J → H → E → F → I

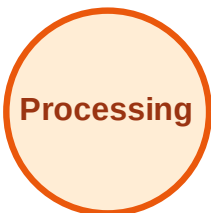
Visited: A, B, C



BFS Traversal

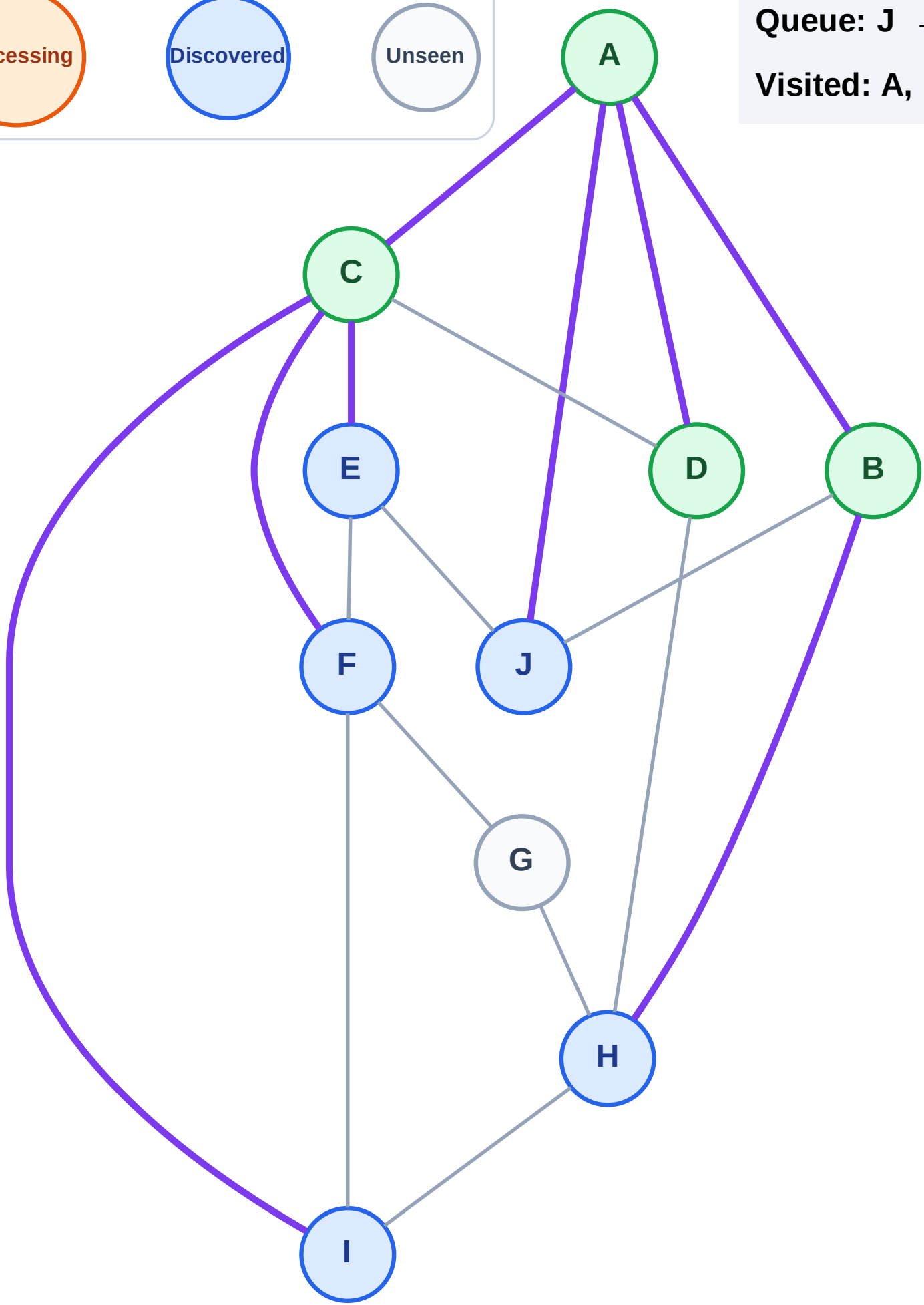
Step 9: No new neighbors from D

Legend



Queue: J → H → E → F → I

Visited: A, B, C, D



BFS Traversal

Step 10: Process node J

Legend

Complete

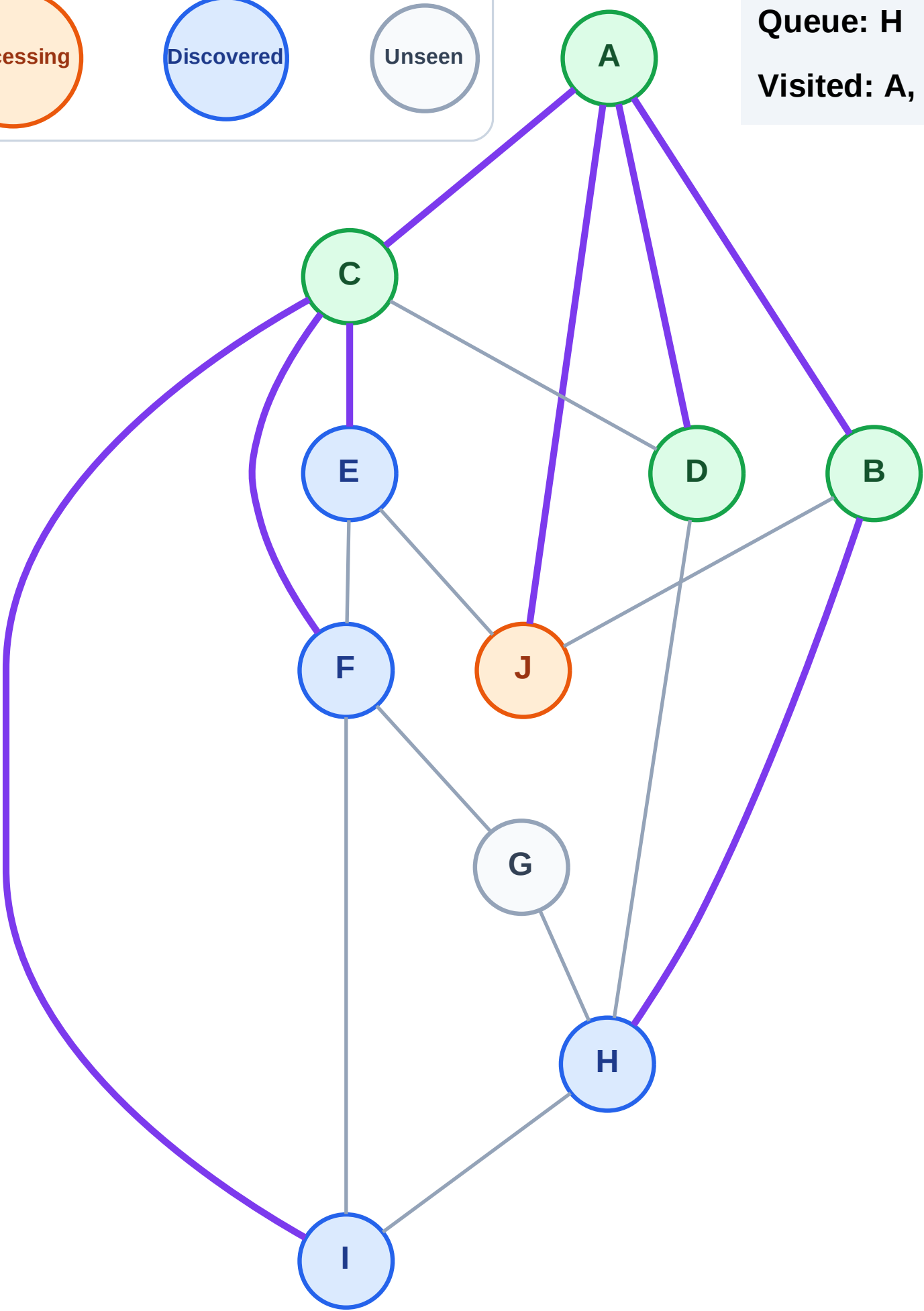
Processing

Discovered

Unseen

Queue: H → E → F → I

Visited: A, B, C, D



BFS Traversal

Step 11: No new neighbors from J

Legend

Complete

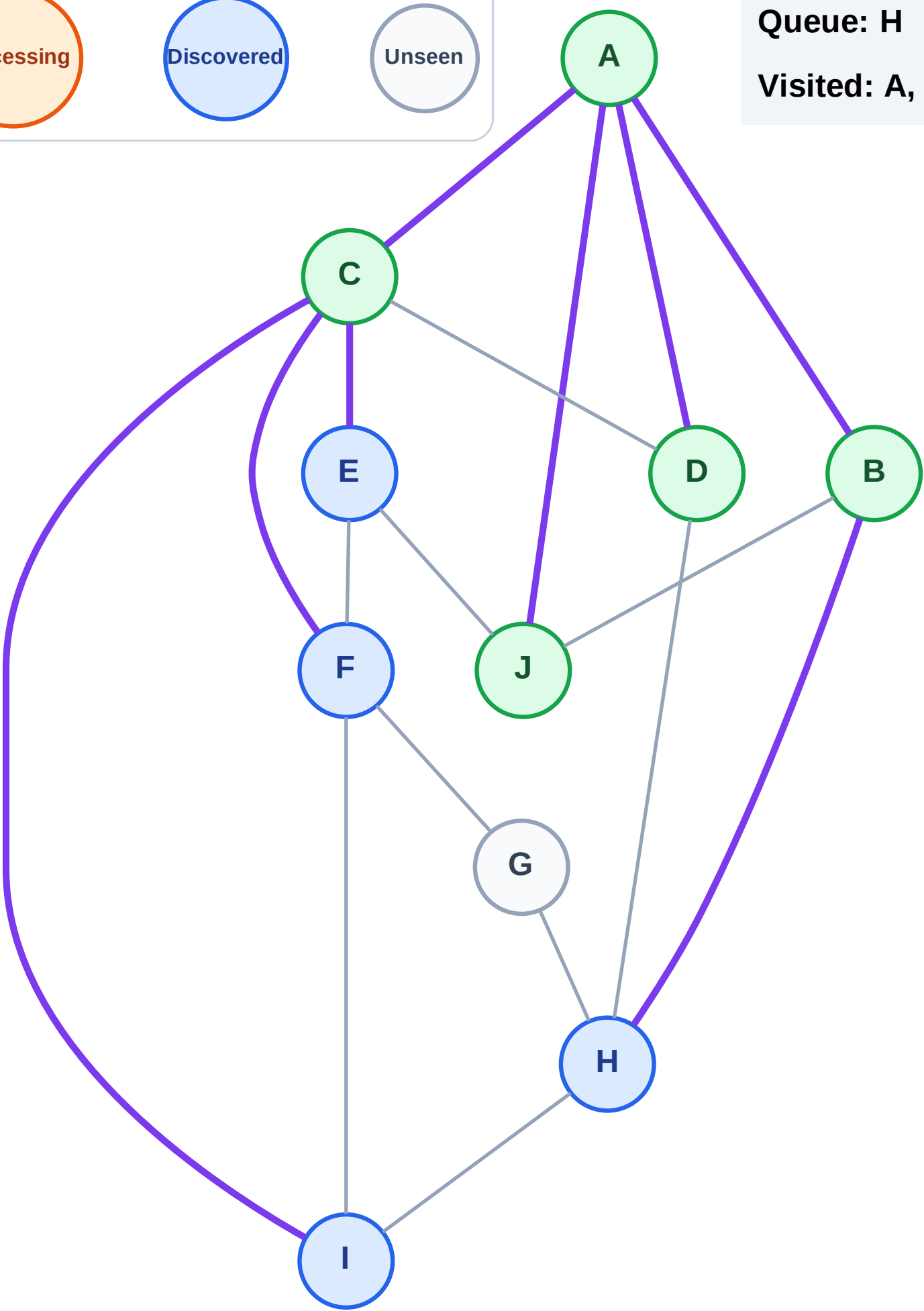
Processing

Discovered

Unseen

Queue: H → E → F → I

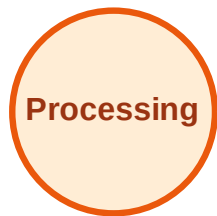
Visited: A, B, C, D, J



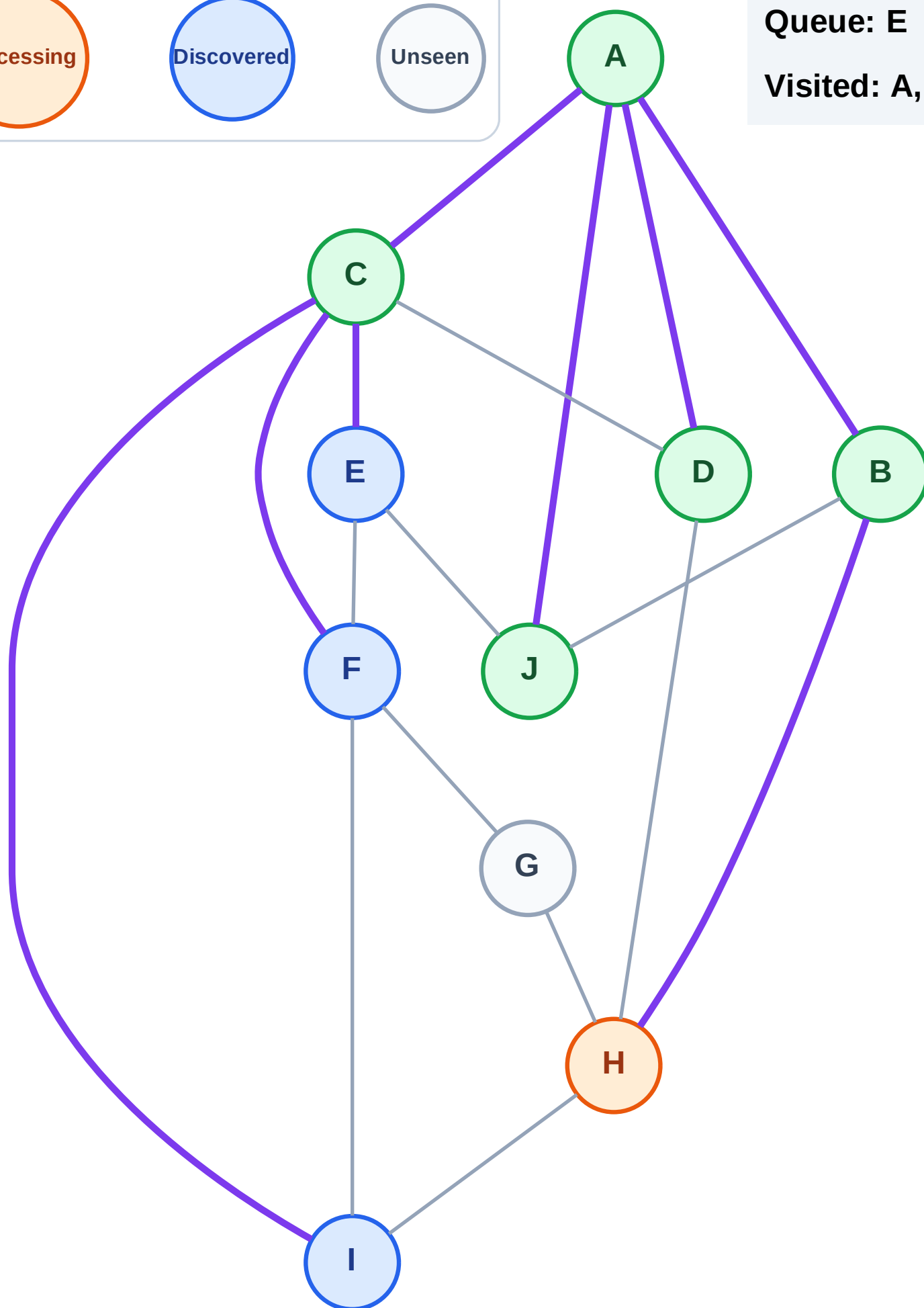
BFS Traversal

Step 12: Process node H

Legend

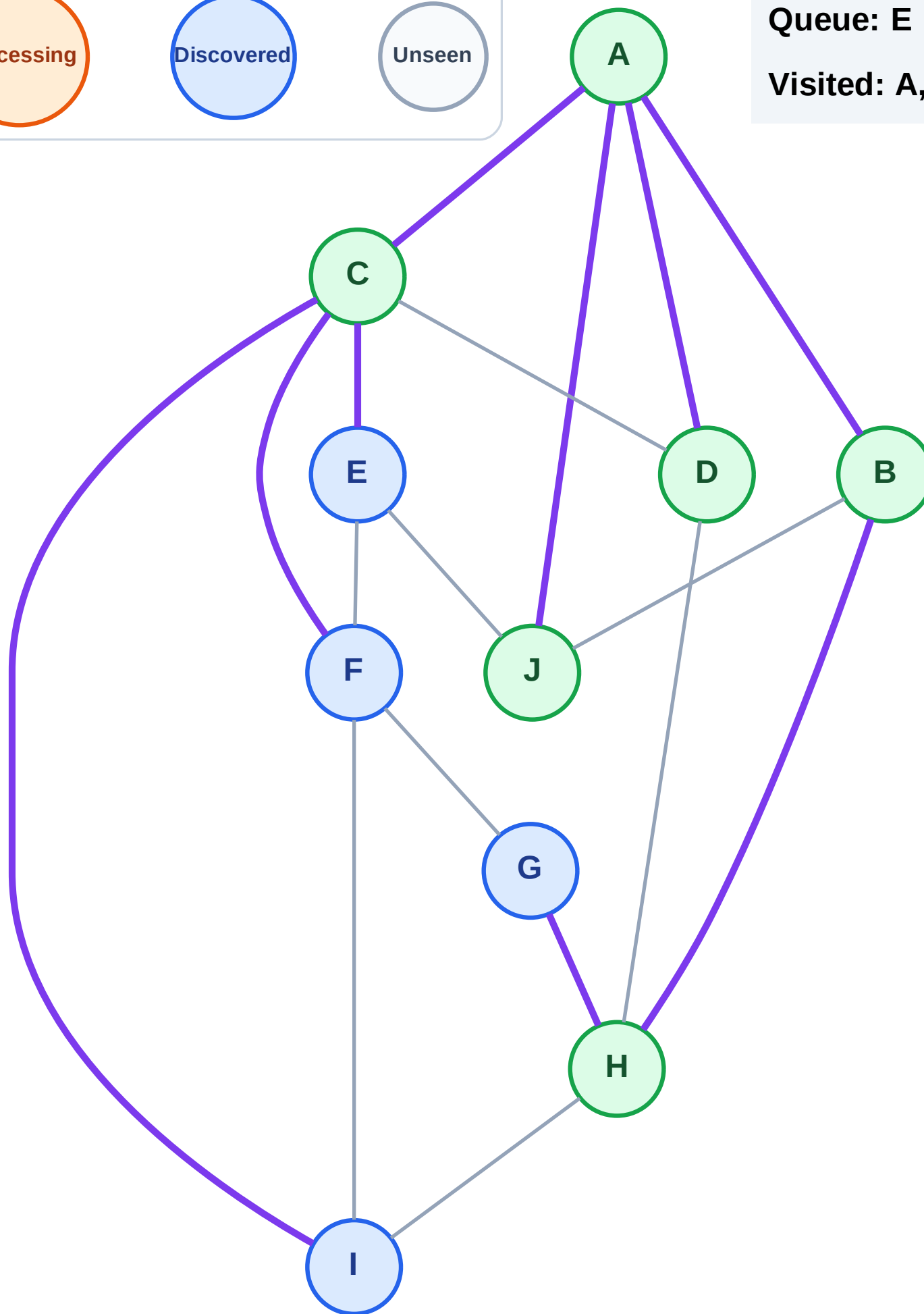
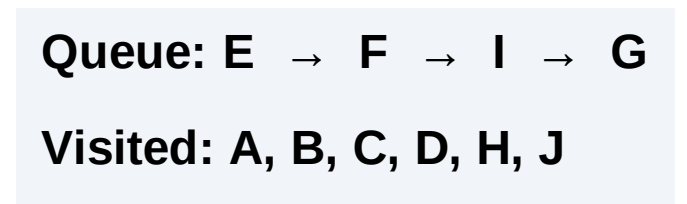


Queue: E → F → I
Visited: A, B, C, D, J



Step 13: Discover G from H

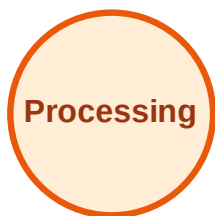
Step 13: Discover G from H



BFS Traversal

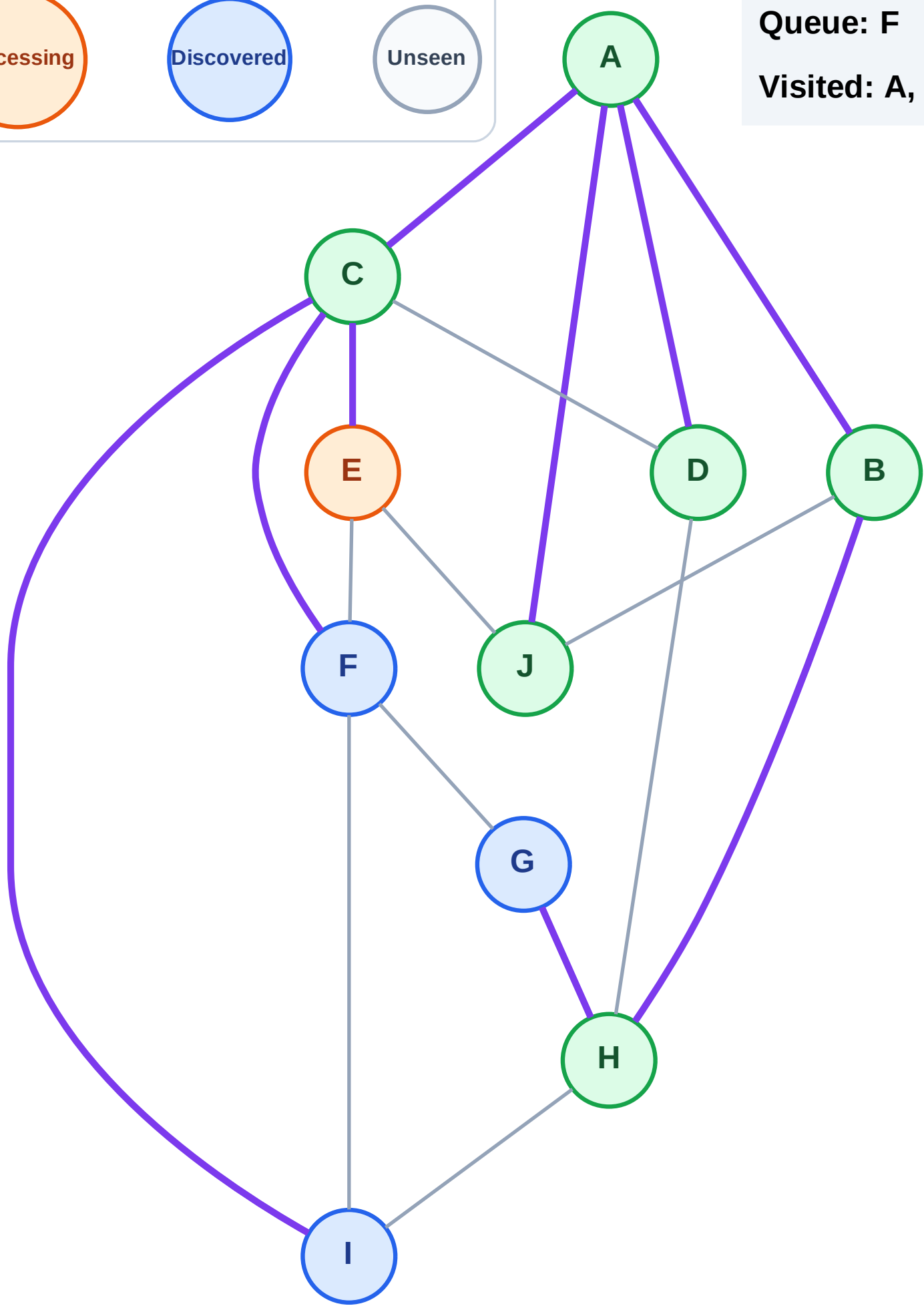
Step 14: Process node E

Legend



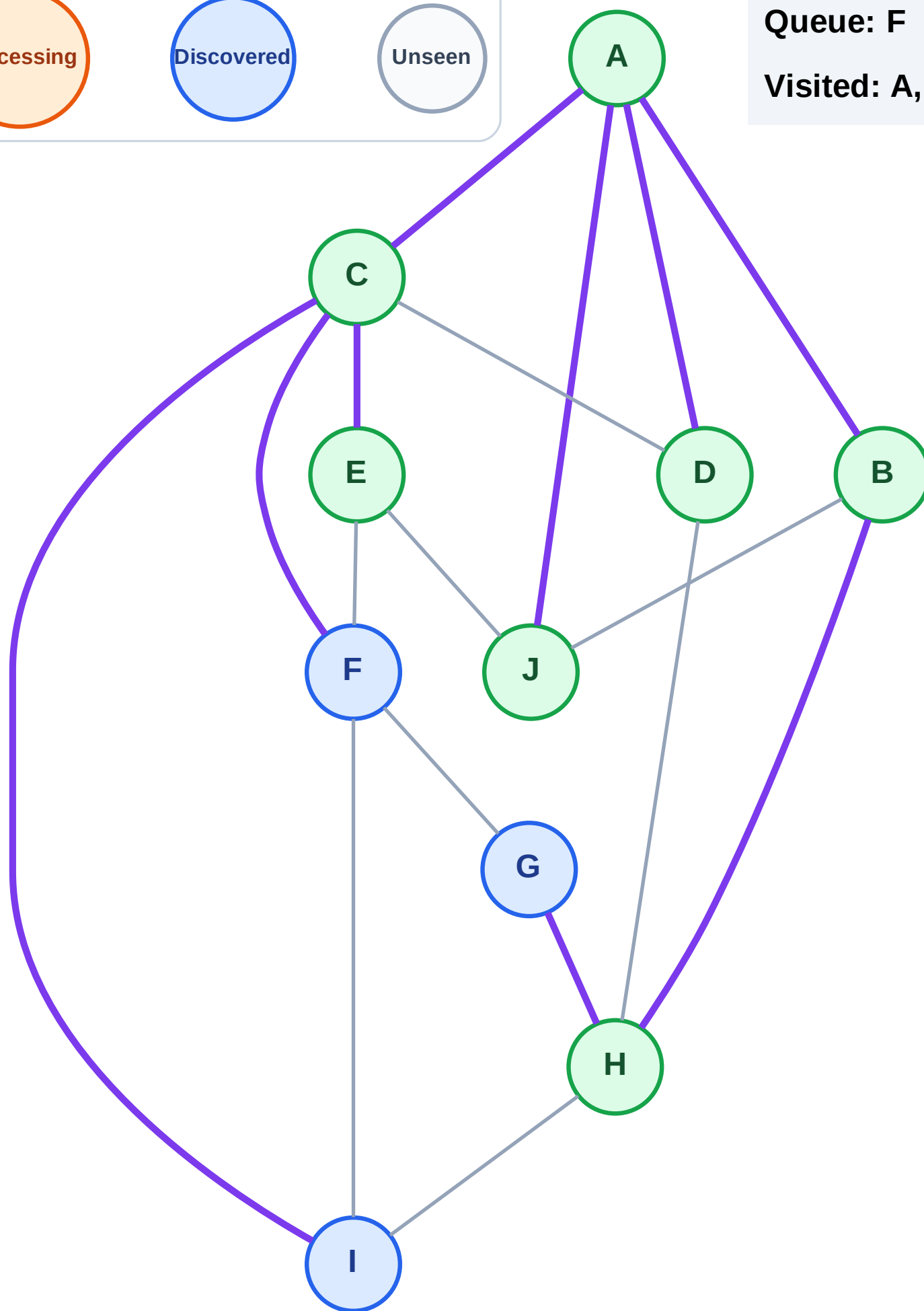
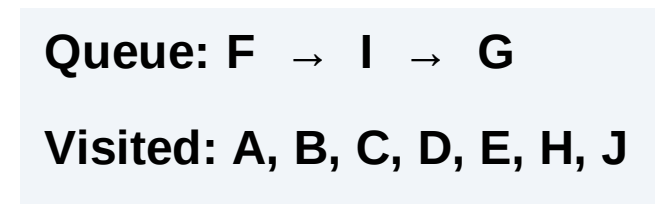
Queue: F → I → G

Visited: A, B, C, D, H, J



Step 15: No new neighbors from E

Step 15: No new neighbors from E



BFS Traversal

Step 16: Process node F

Legend

Complete

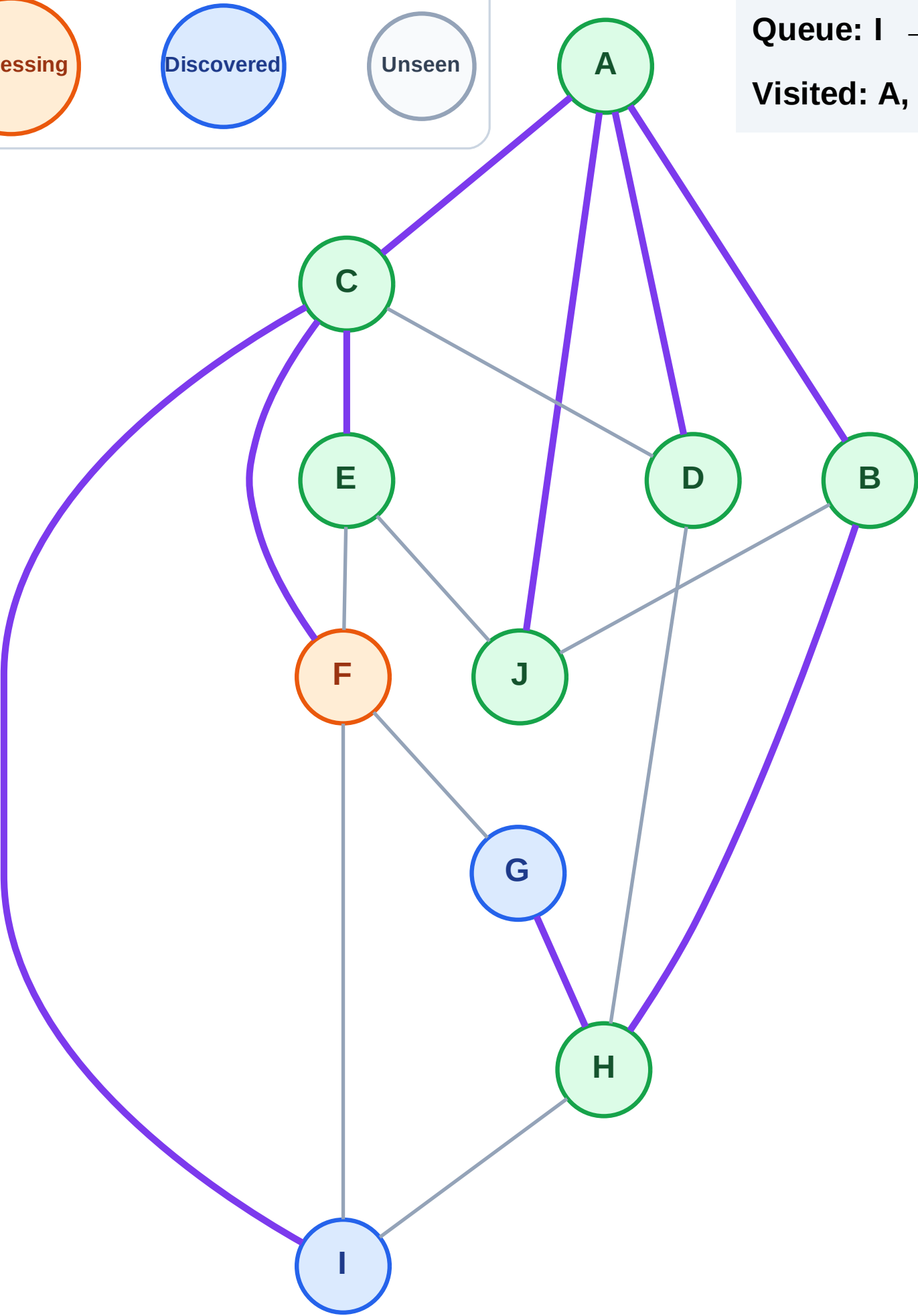
Processing

Discovered

Unseen

Queue: I → G

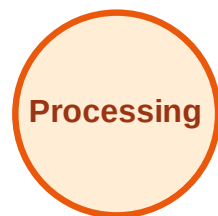
Visited: A, B, C, D, E, H, J



BFS Traversal

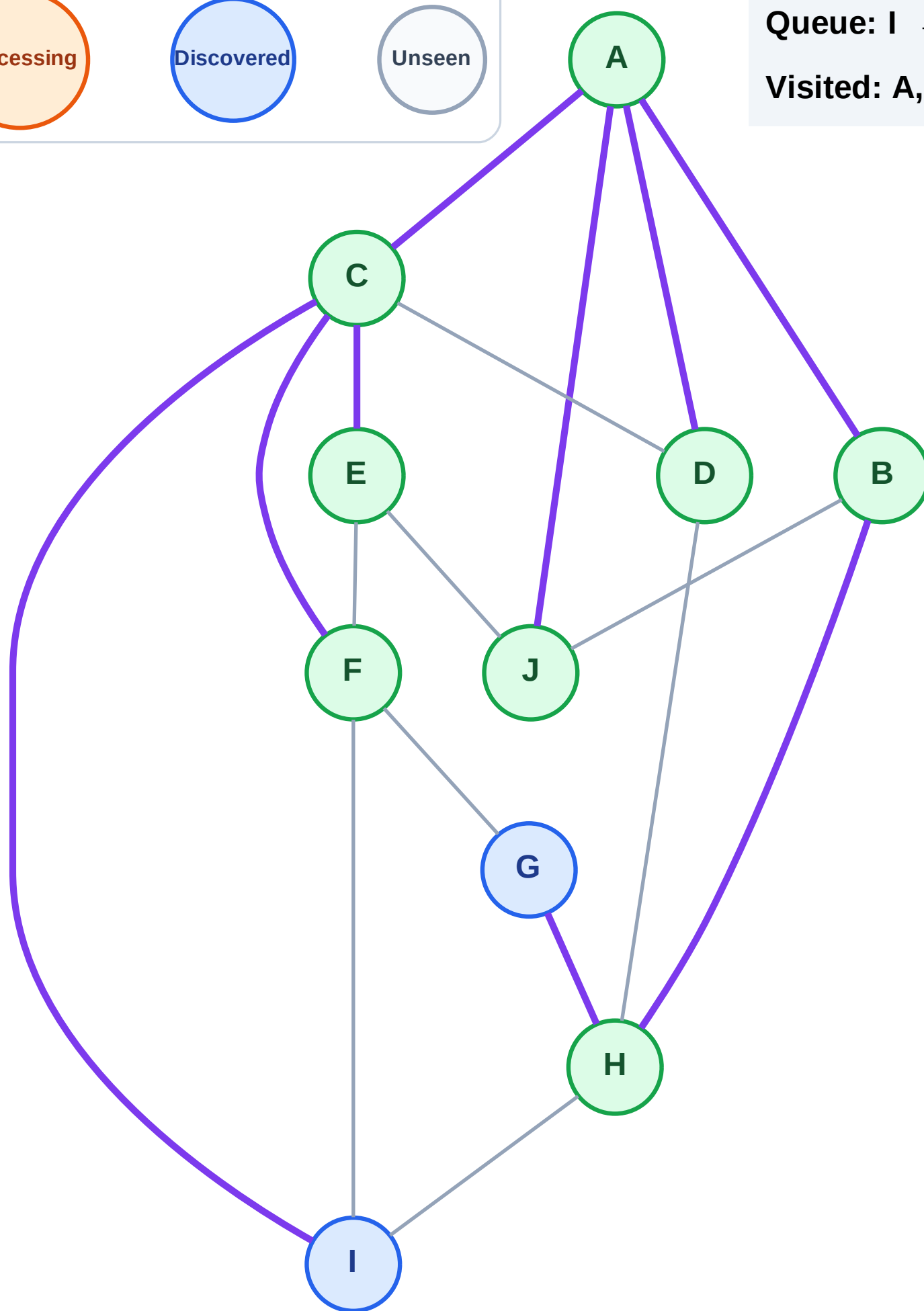
Step 17: No new neighbors from F

Legend



Queue: I → G

Visited: A, B, C, D, E, F, H, J



BFS Traversal

Step 18: Process node I

Legend

Complete

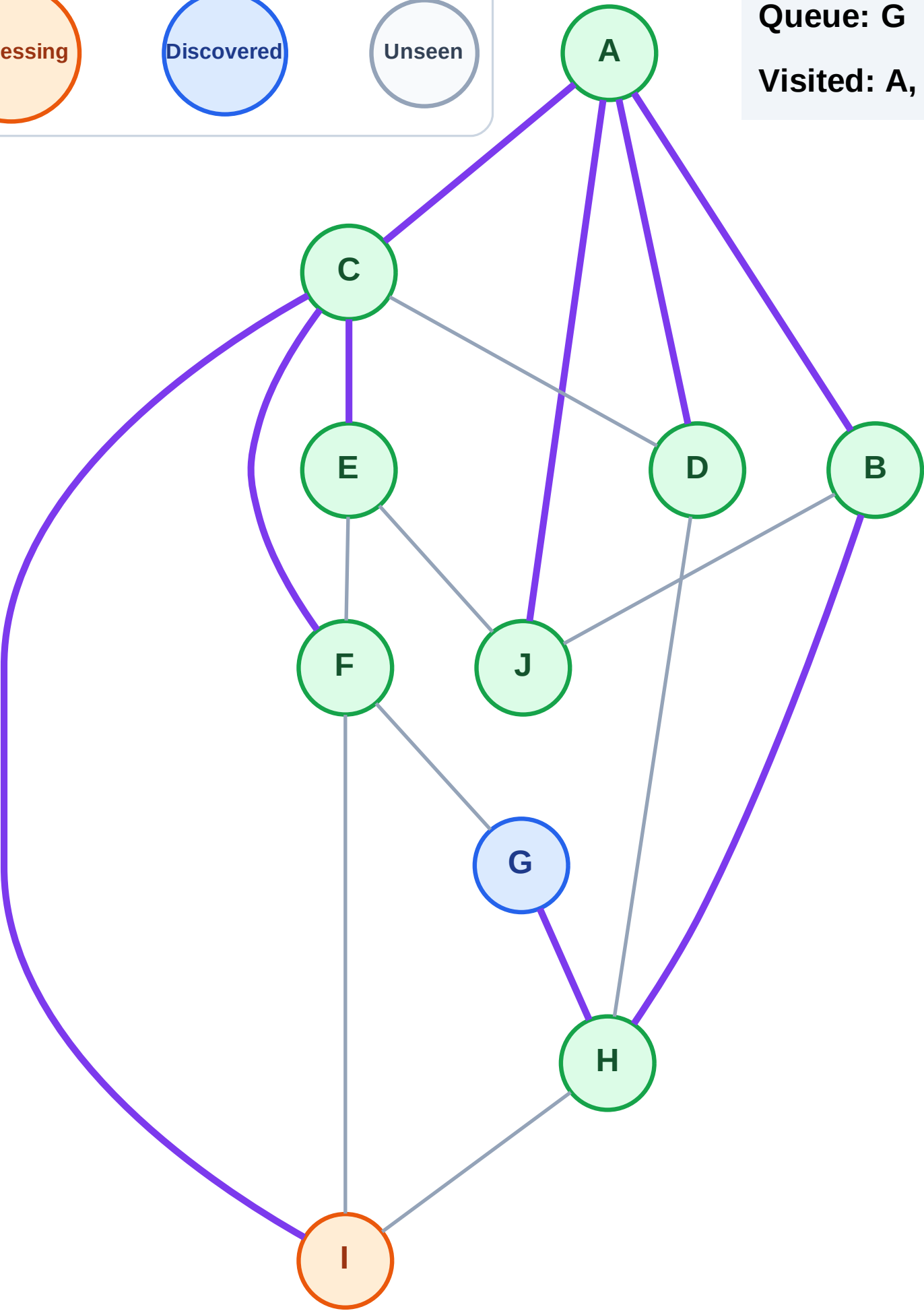
Processing

Discovered

Unseen

Queue: G

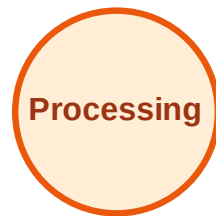
Visited: A, B, C, D, E, F, H, J



BFS Traversal

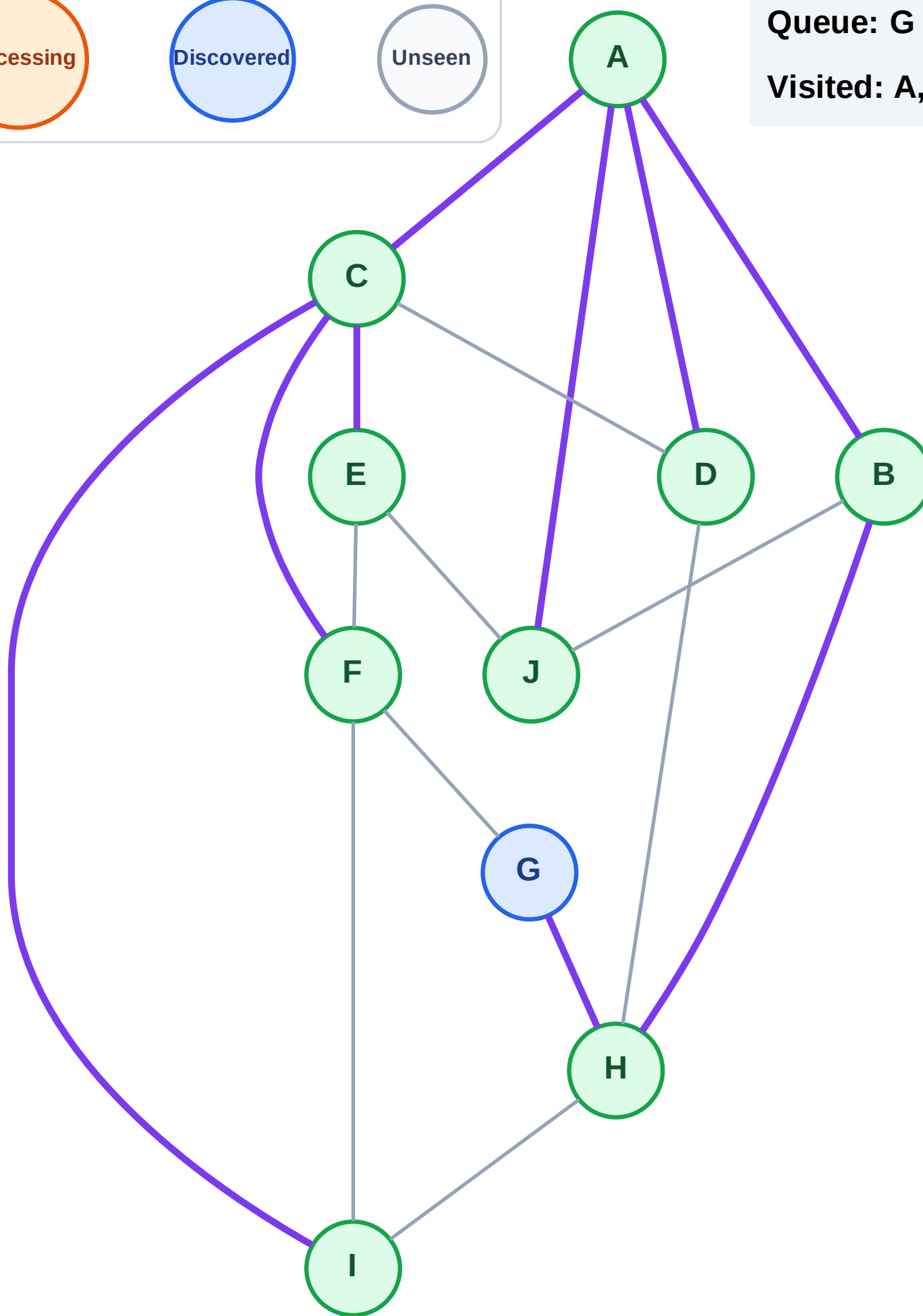
Step 19: No new neighbors from I

Legend



Queue: G

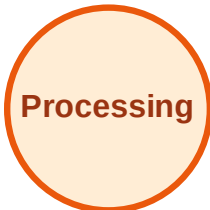
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

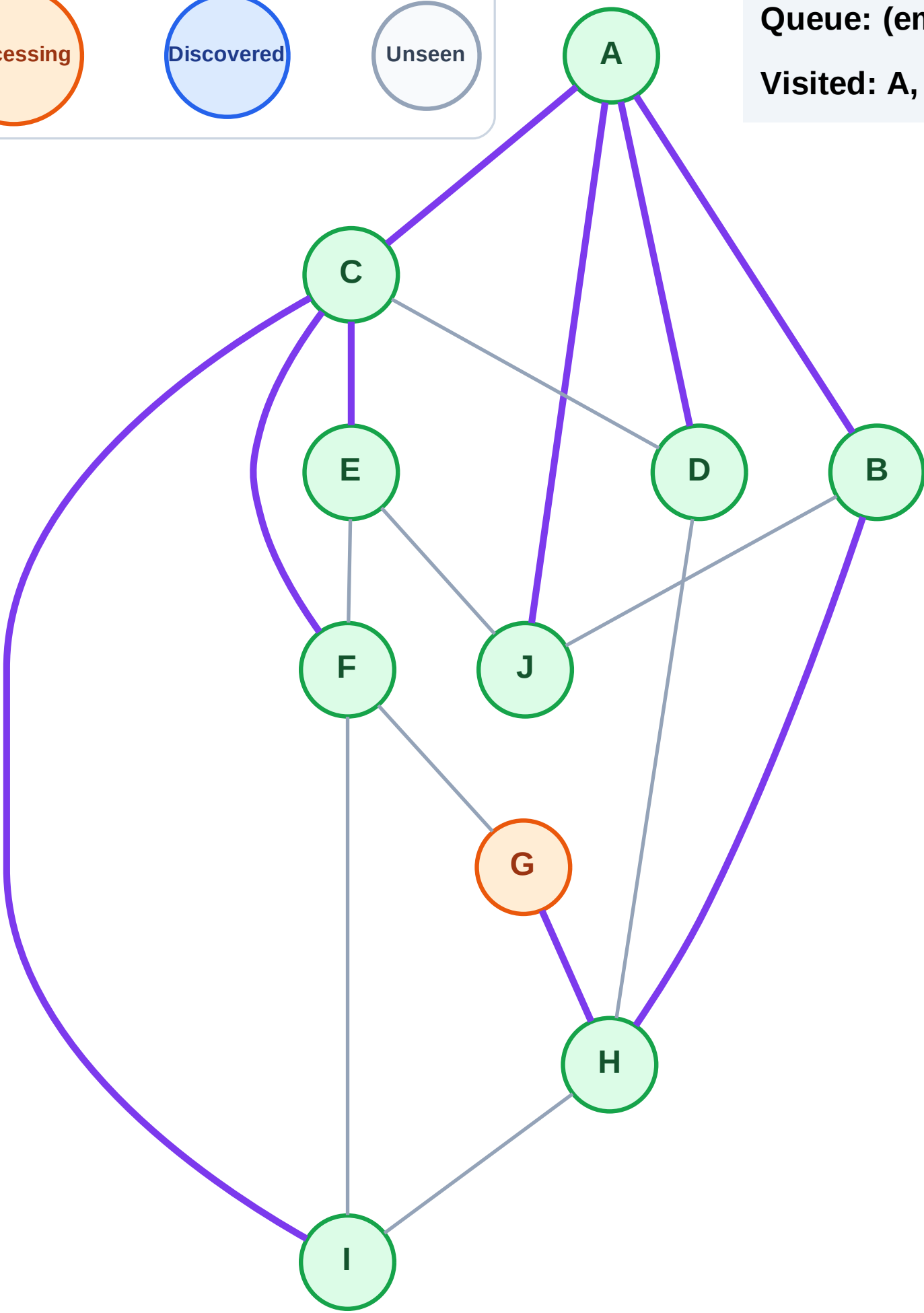
Step 20: Process node G

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

Step 21: No new neighbors from G

Legend



Complete



Processing



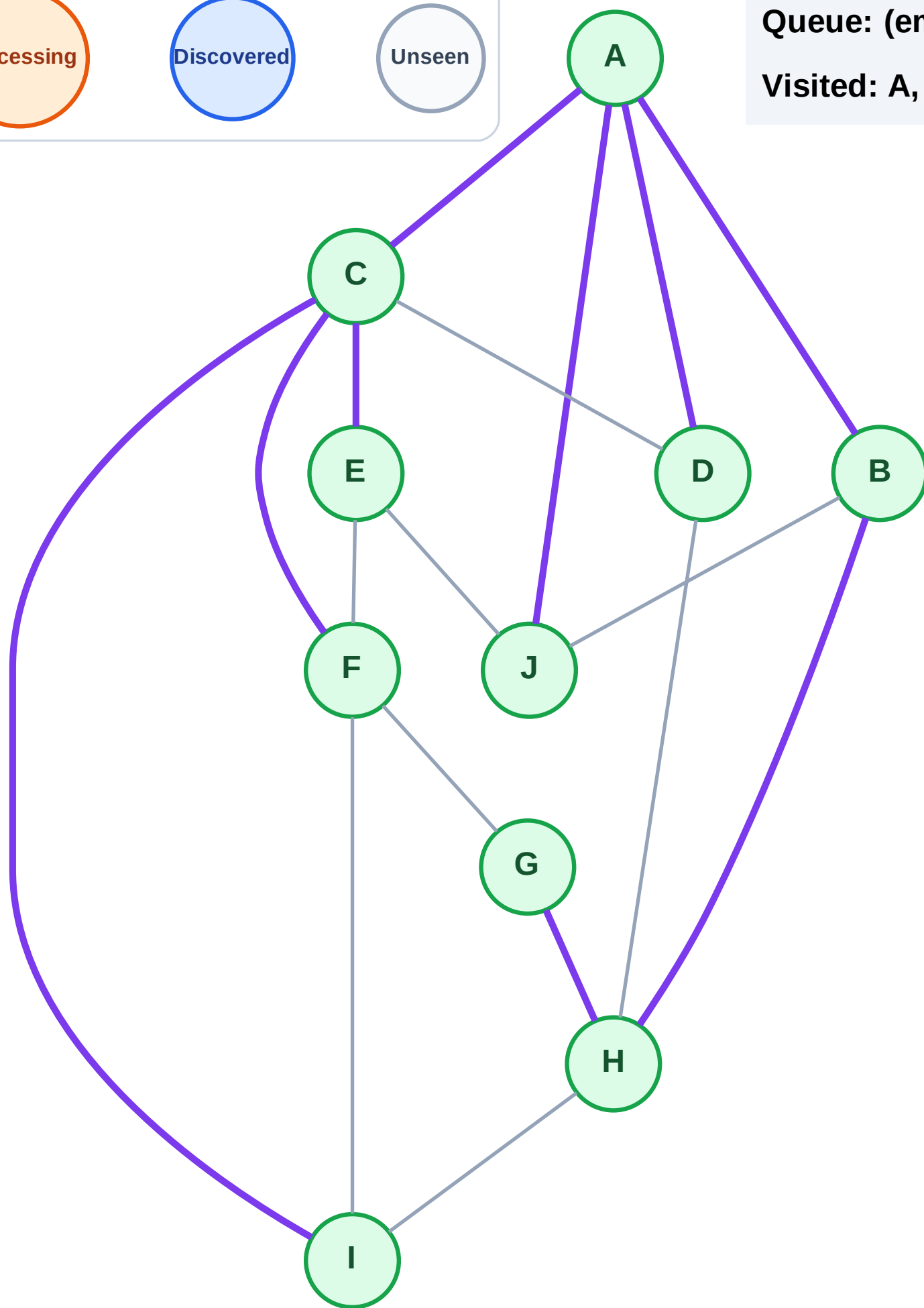
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

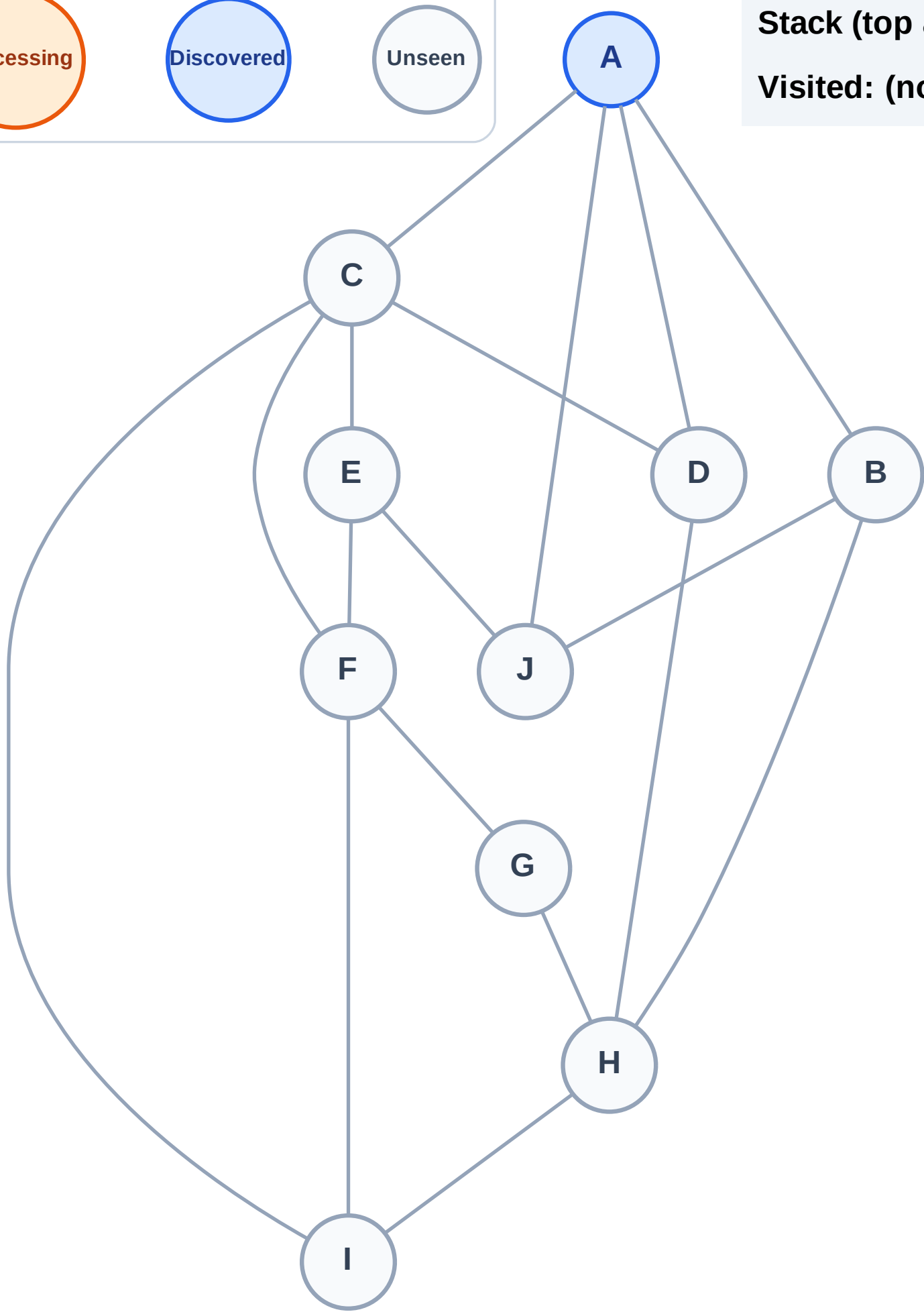
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

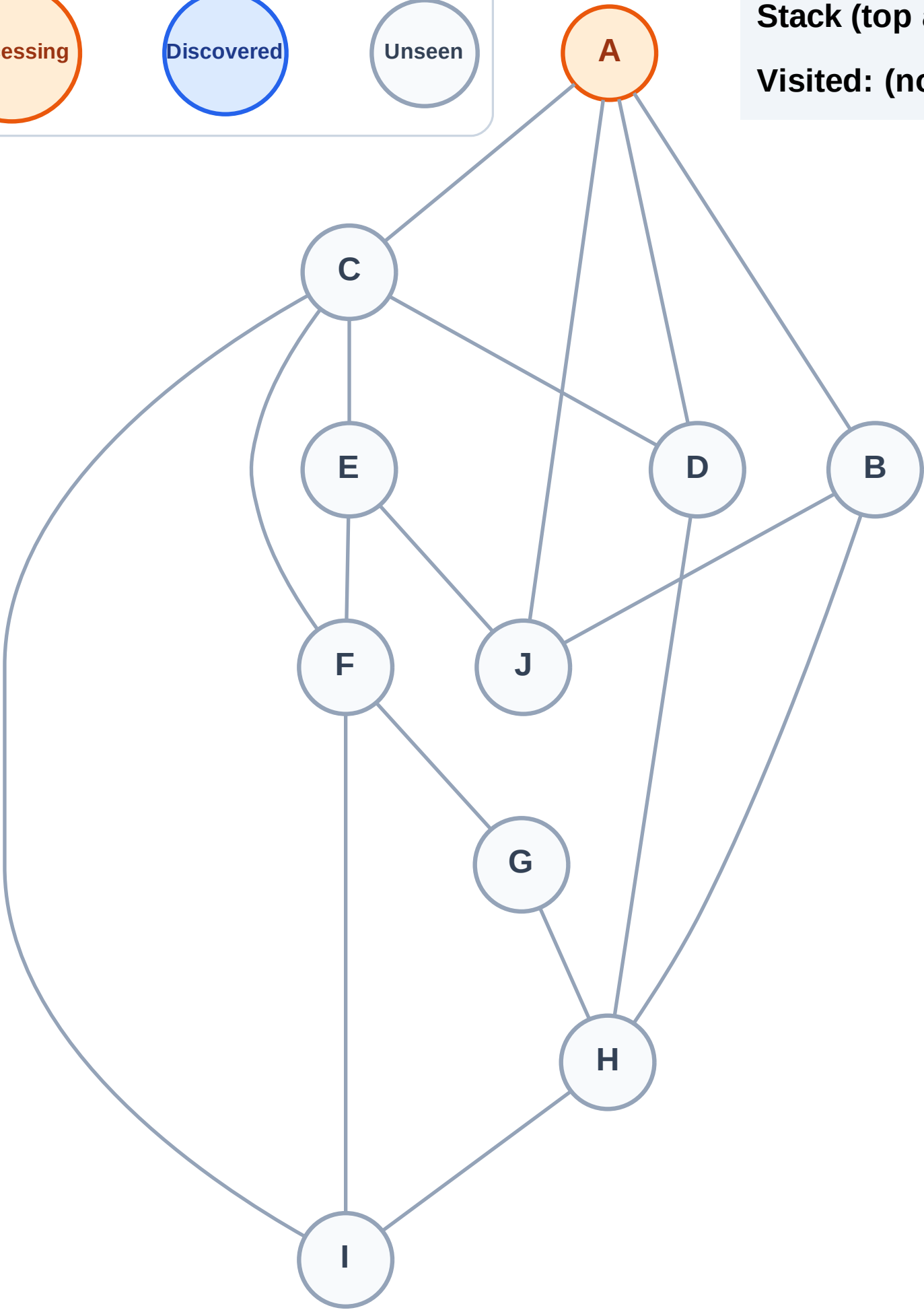
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover J, D, C, B from A

Legend

Complete

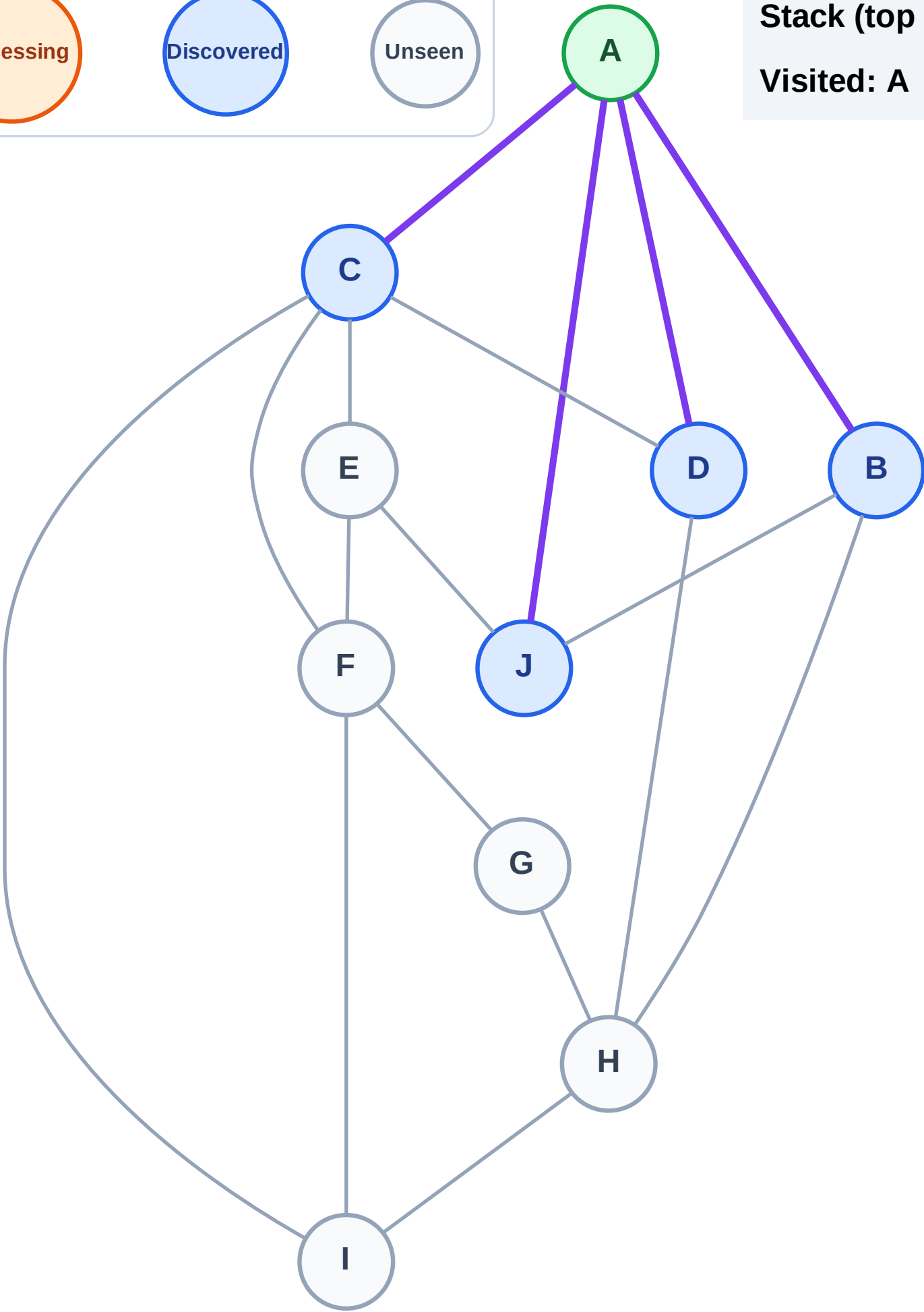
Processing

Discovered

Unseen

Stack (top at right): J | D | C | B

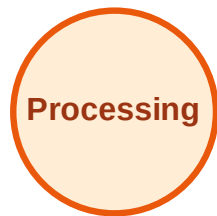
Visited: A



DFS Traversal

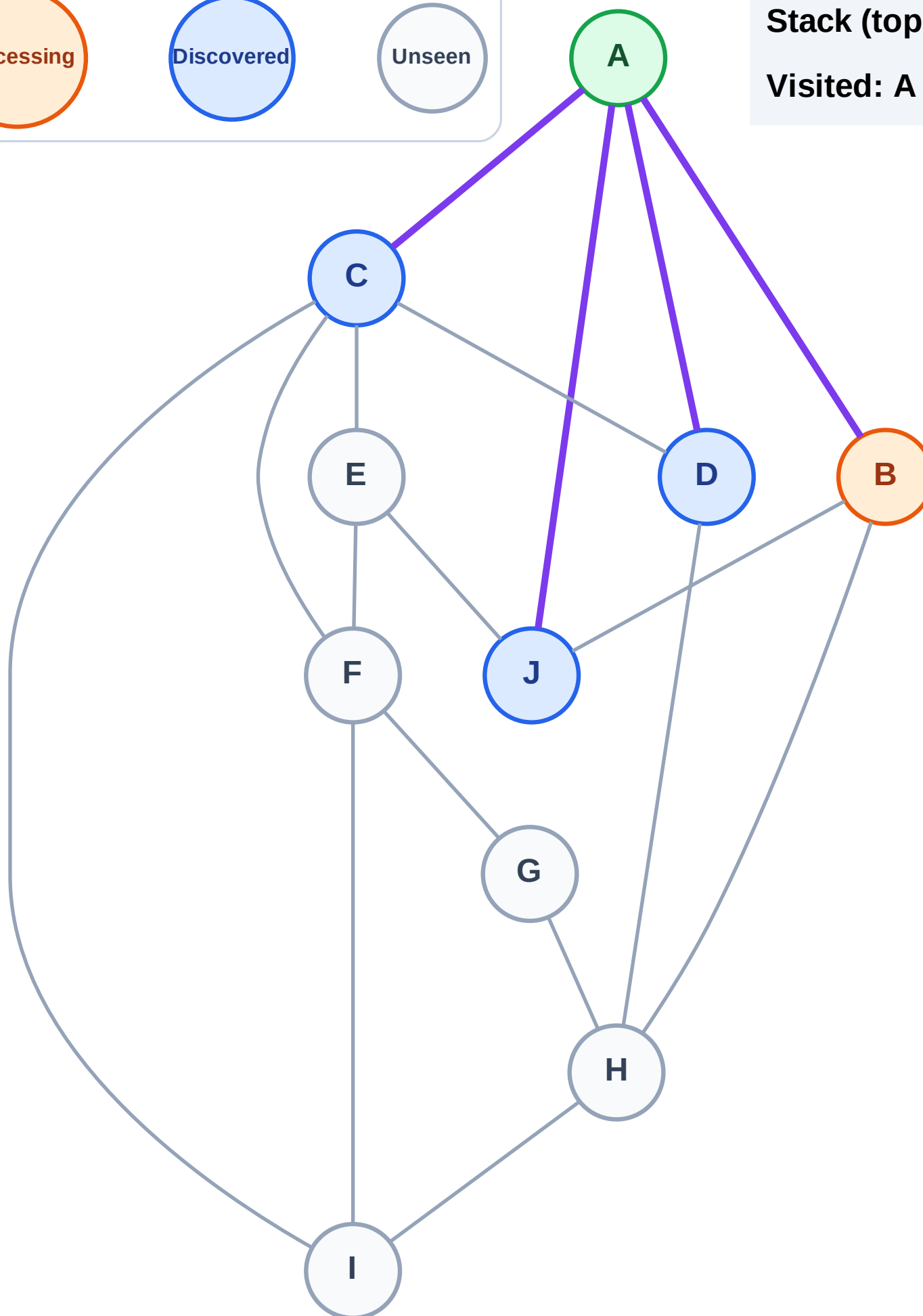
Step 4: Process node B

Legend



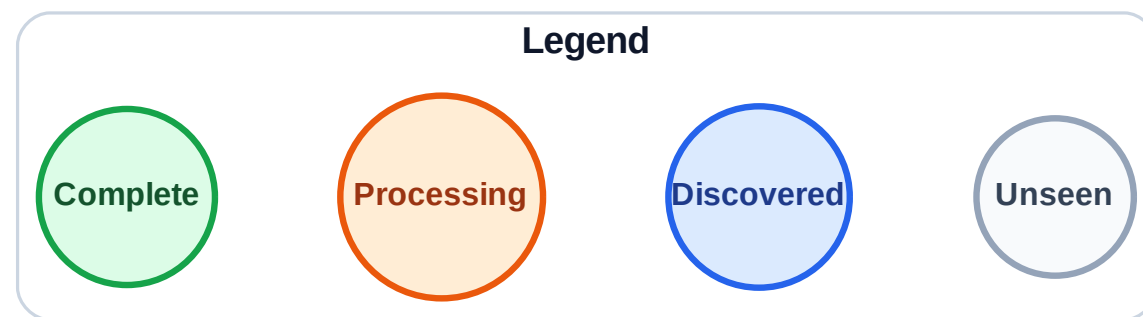
Stack (top at right): J | D | C

Visited: A

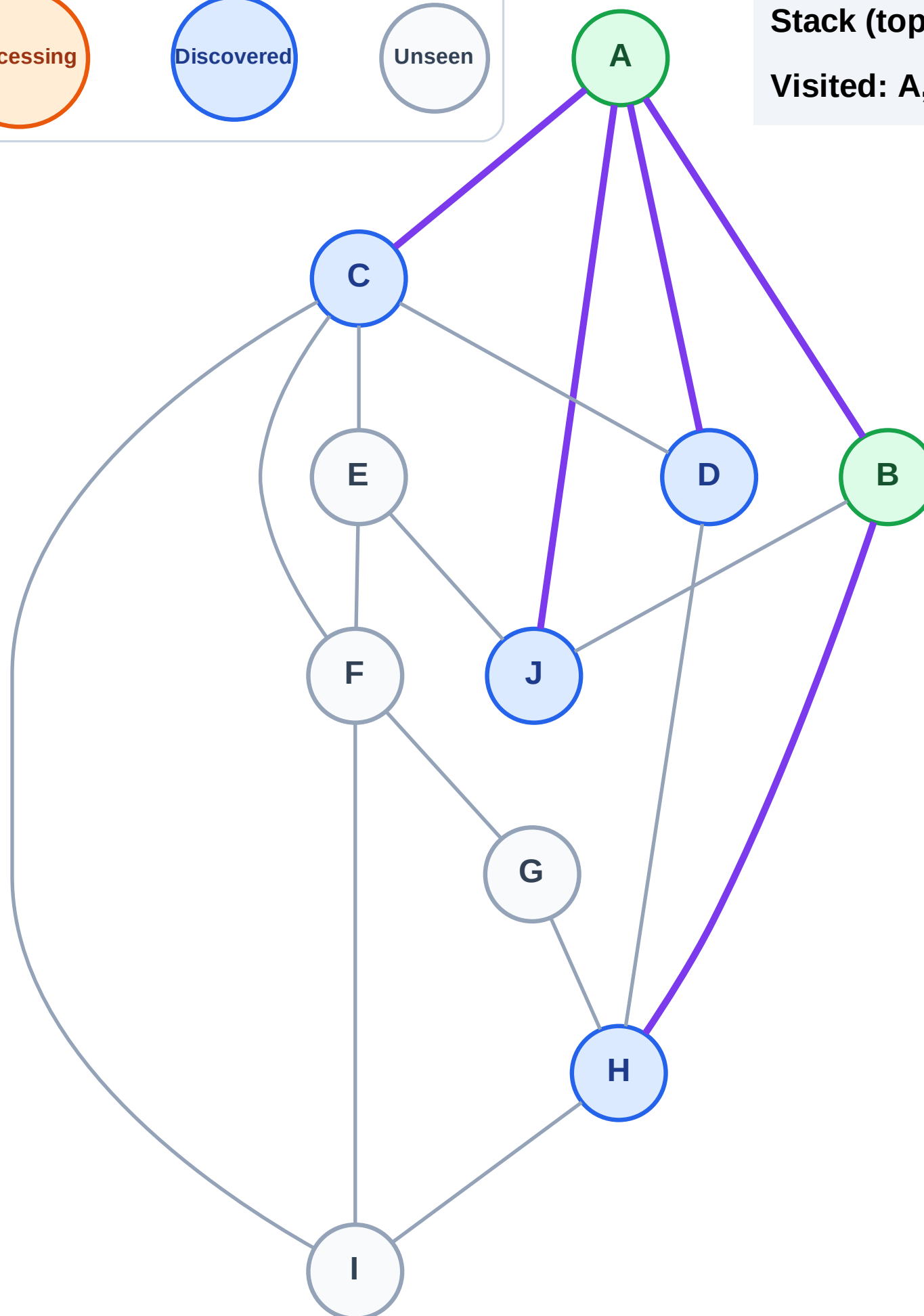


DFS Traversal

Step 5: Discover H from B



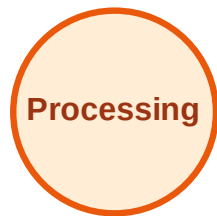
Stack (top at right): J | D | C | H
Visited: A, B



DFS Traversal

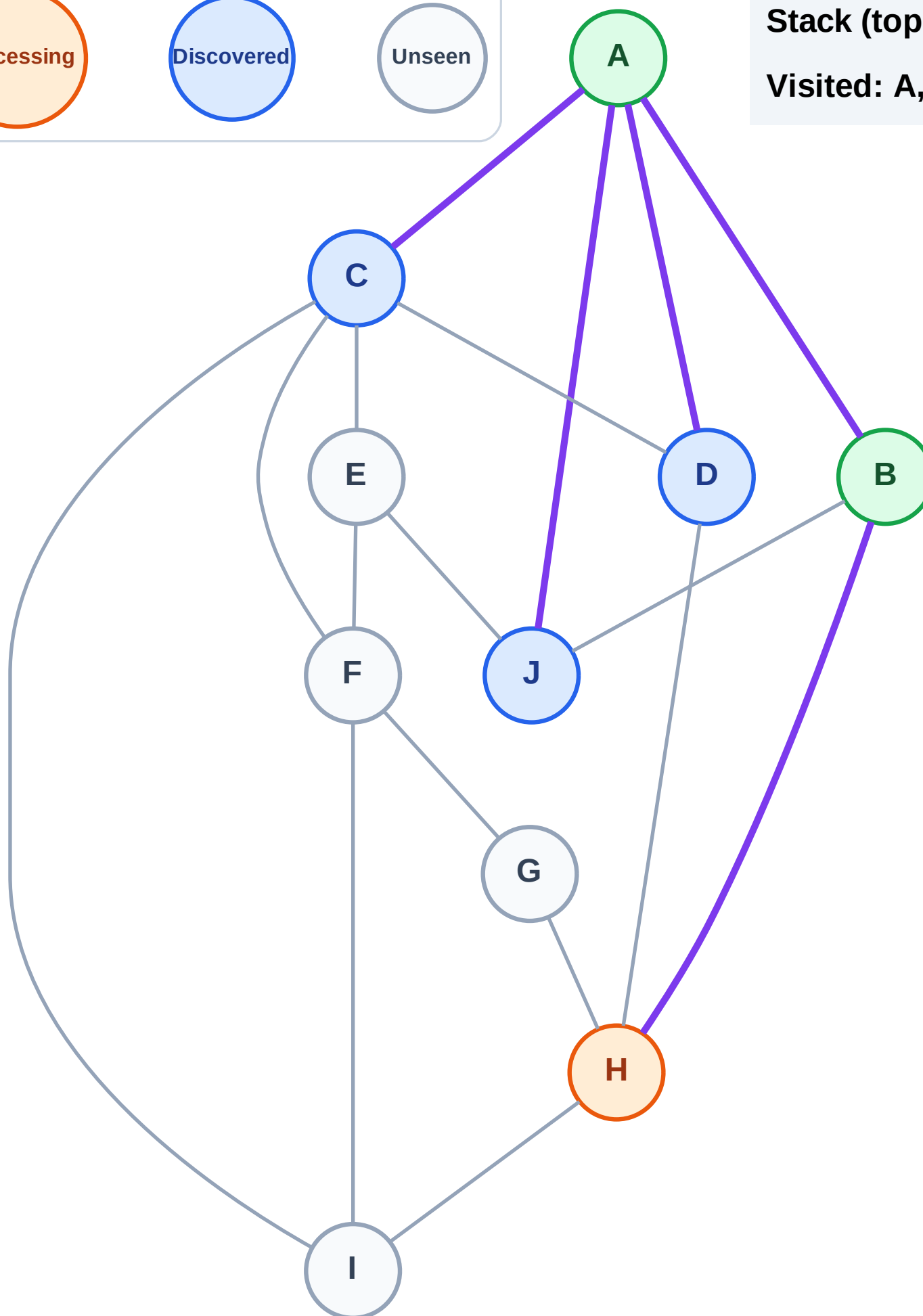
Step 6: Process node H

Legend



Stack (top at right): J | D | C

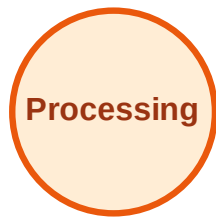
Visited: A, B



DFS Traversal

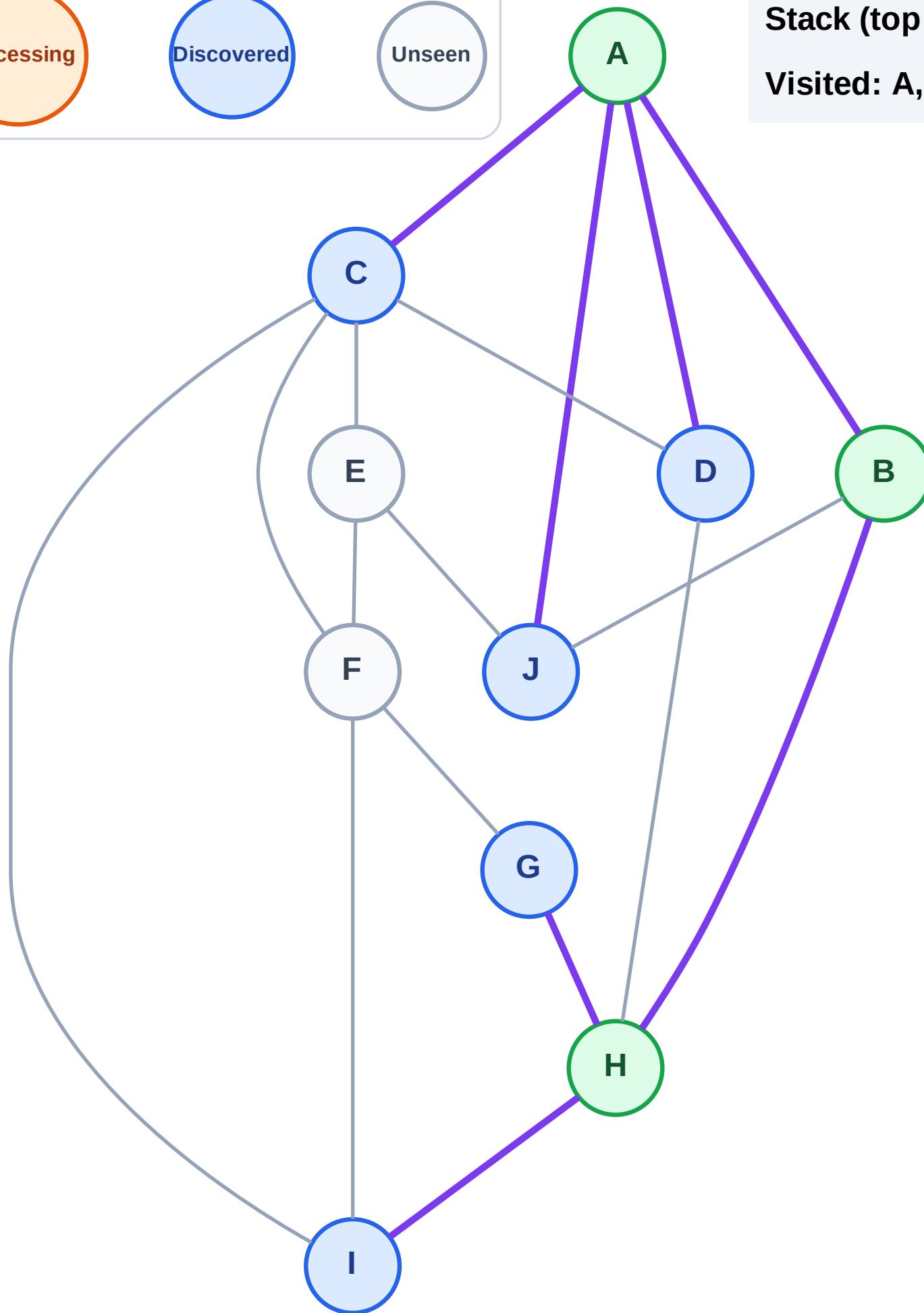
Step 7: Discover I, G from H

Legend



Stack (top at right): J | D | C | I | G

Visited: A, B, H



DFS Traversal

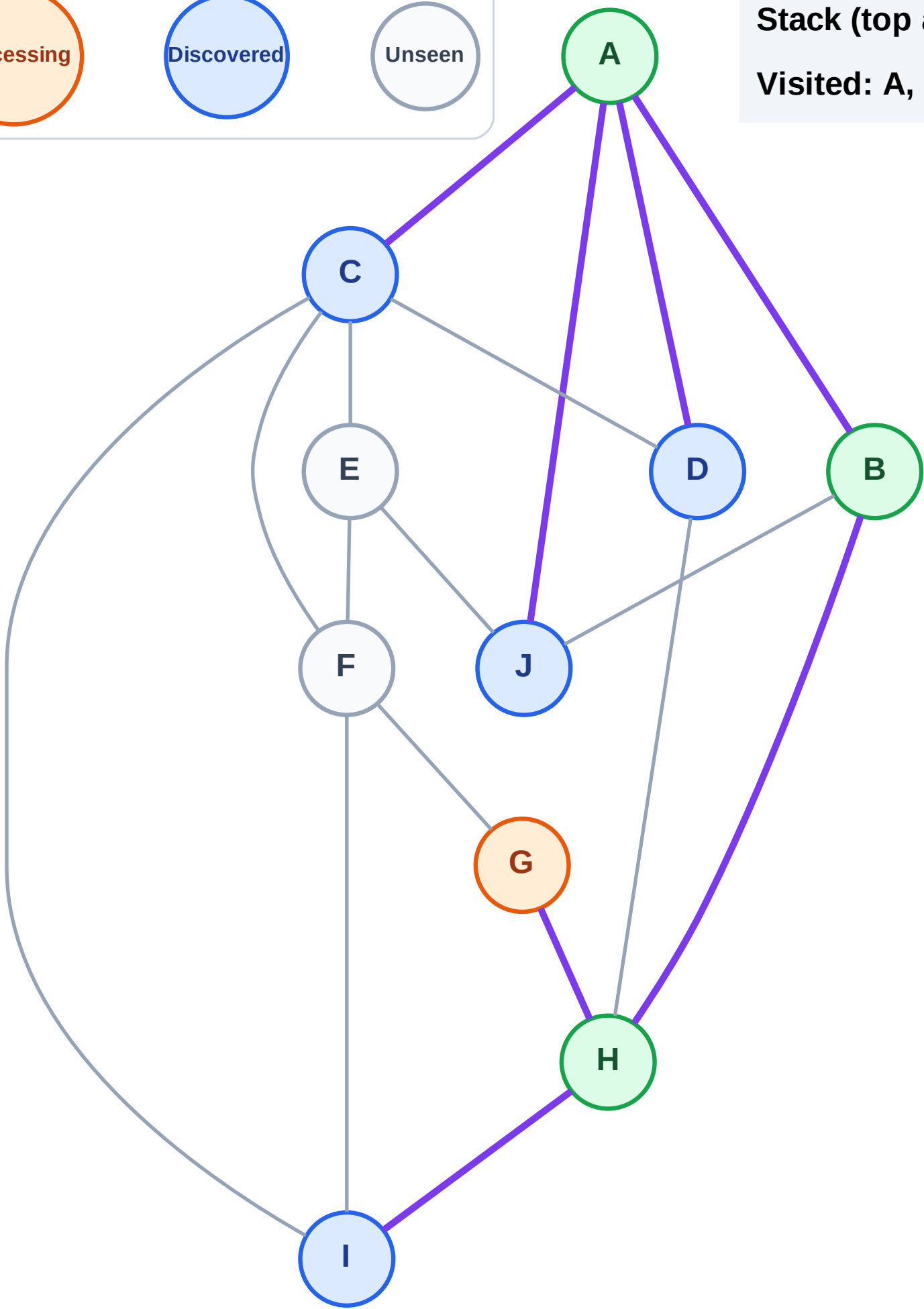
Step 8: Process node G

Legend



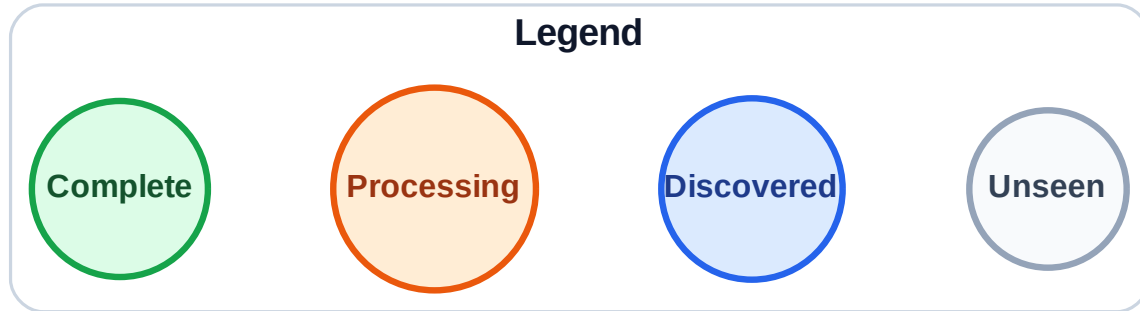
Stack (top at right): J | D | C | I

Visited: A, B, H



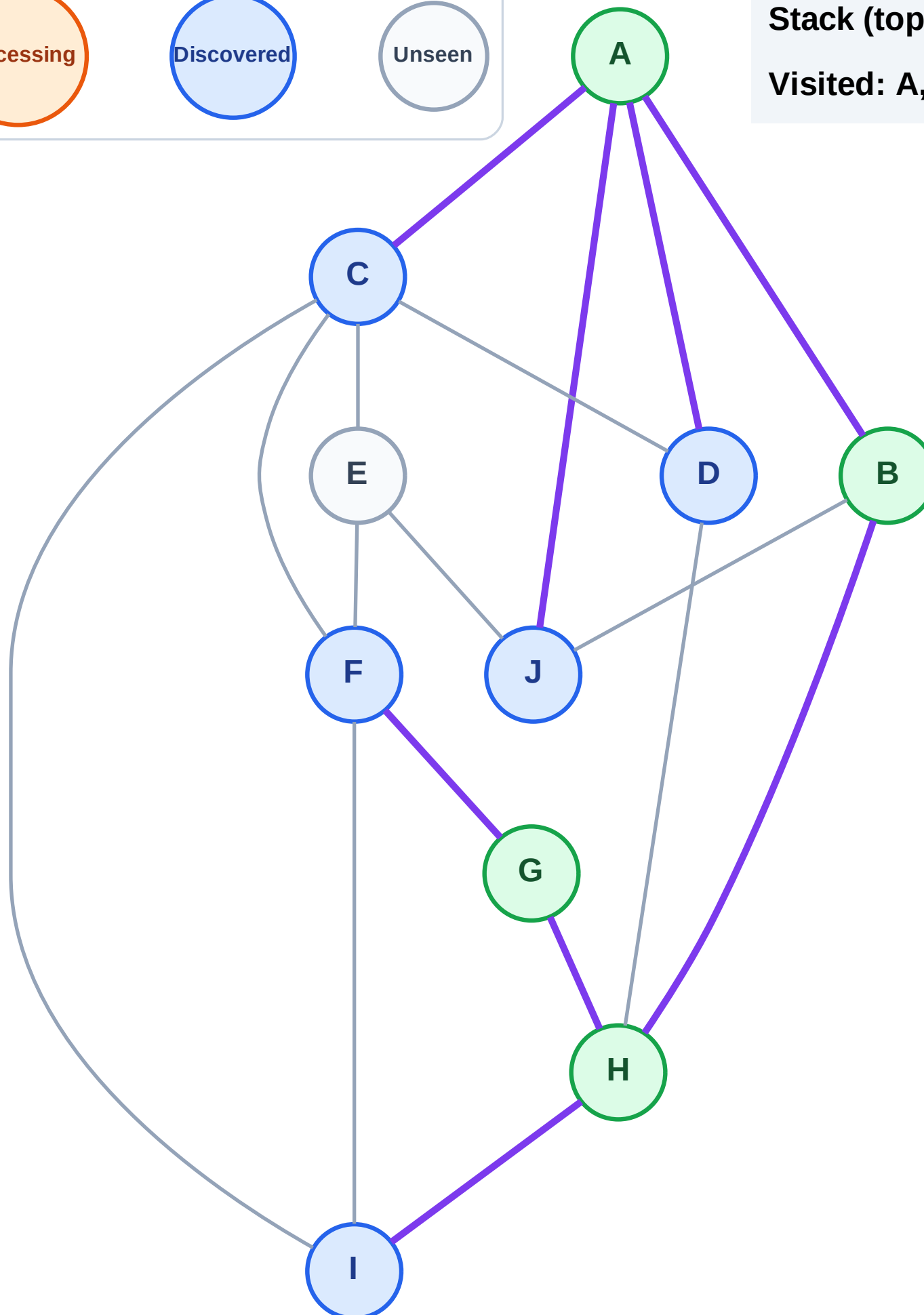
Step 9: Discover F from G

Step 9: Discover F from G

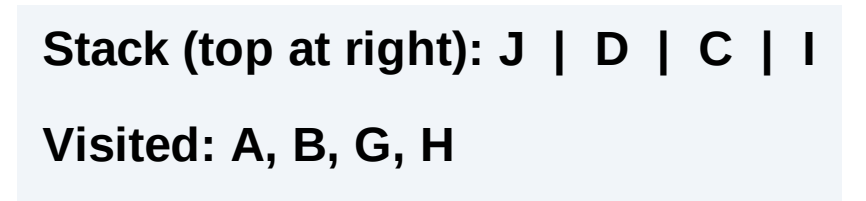


Stack (top at right): J | D | C | I | F

Visited: A, B, G, H

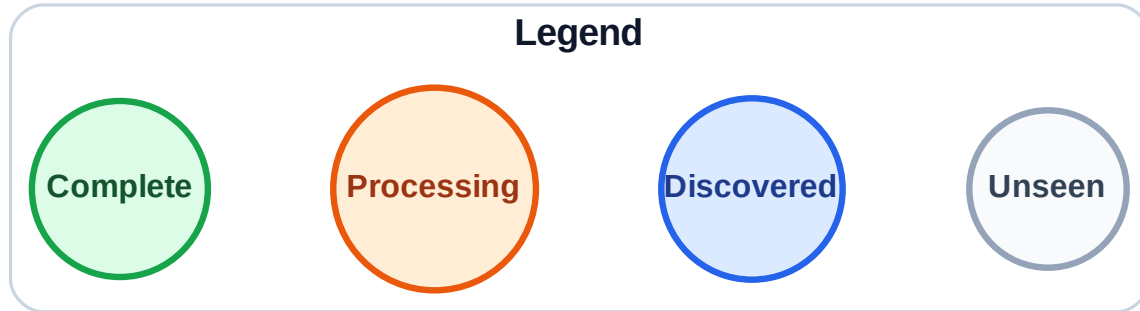


Step 10: Process node F



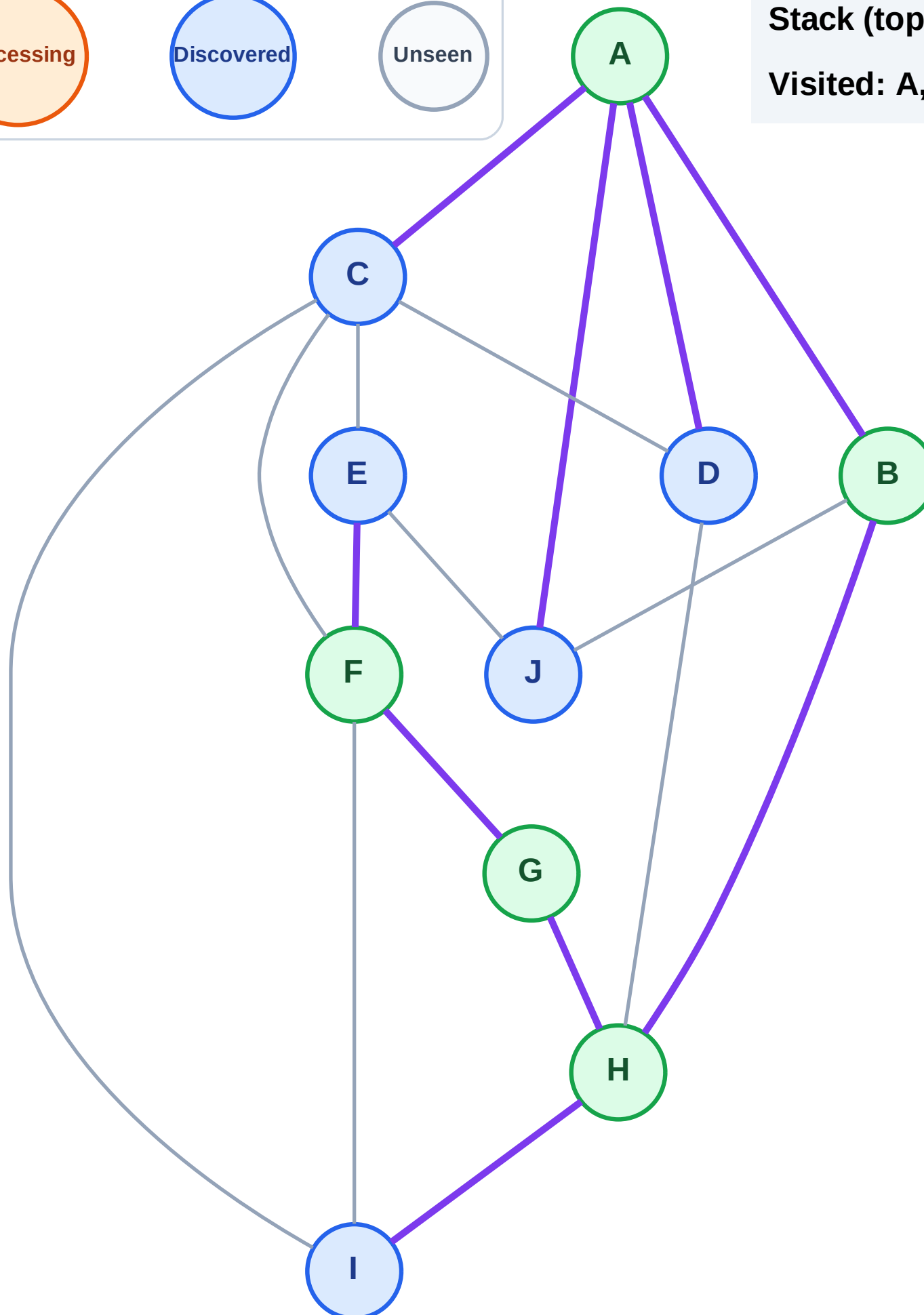
Step 11: Discover E from F

Step 11: Discover E from F



Stack (top at right): J | D | C | I | E

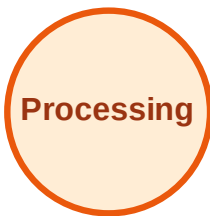
Visited: A, B, F, G, H



DFS Traversal

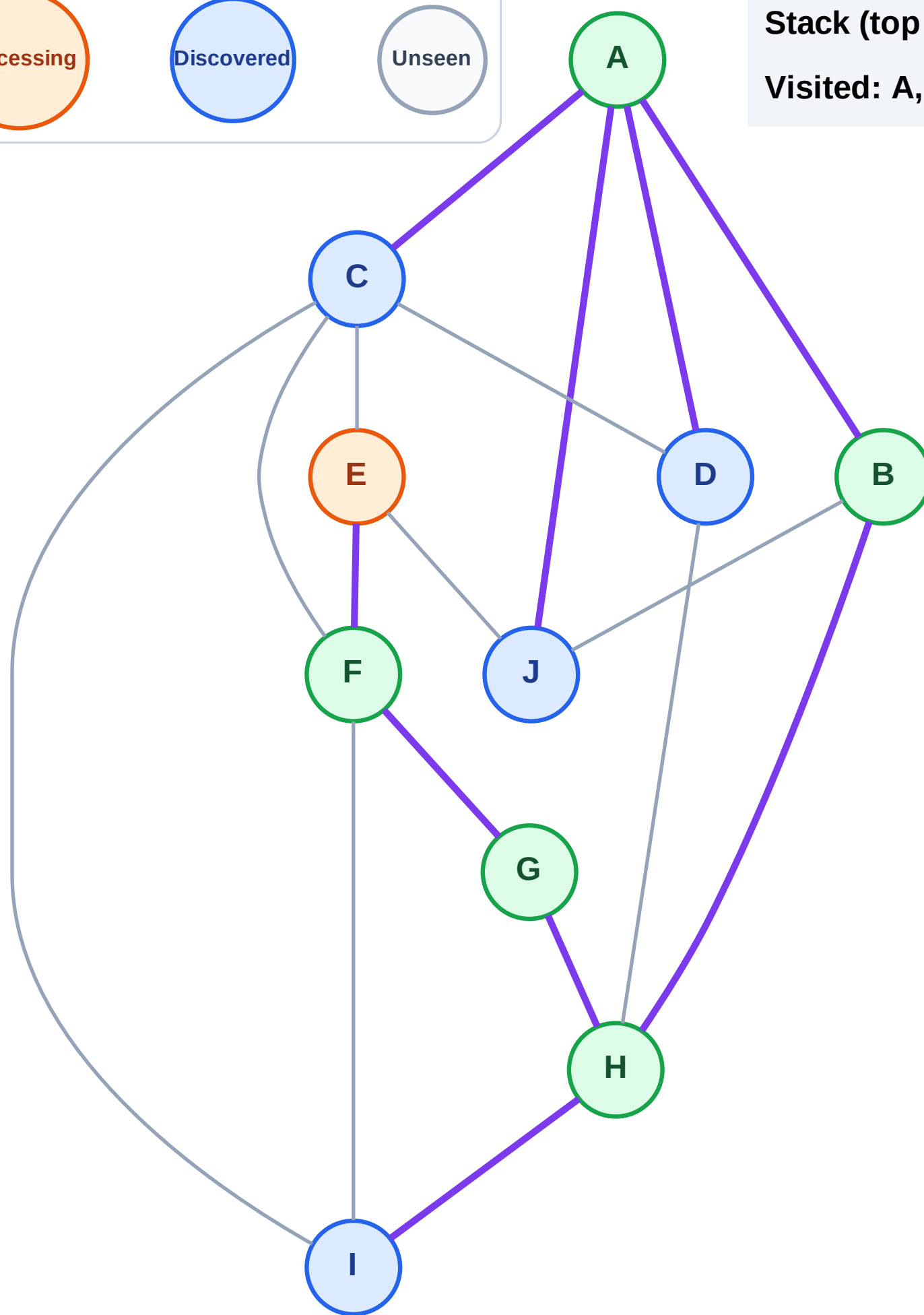
Step 12: Process node E

Legend



Stack (top at right): J | D | C | I

Visited: A, B, F, G, H



DFS Traversal

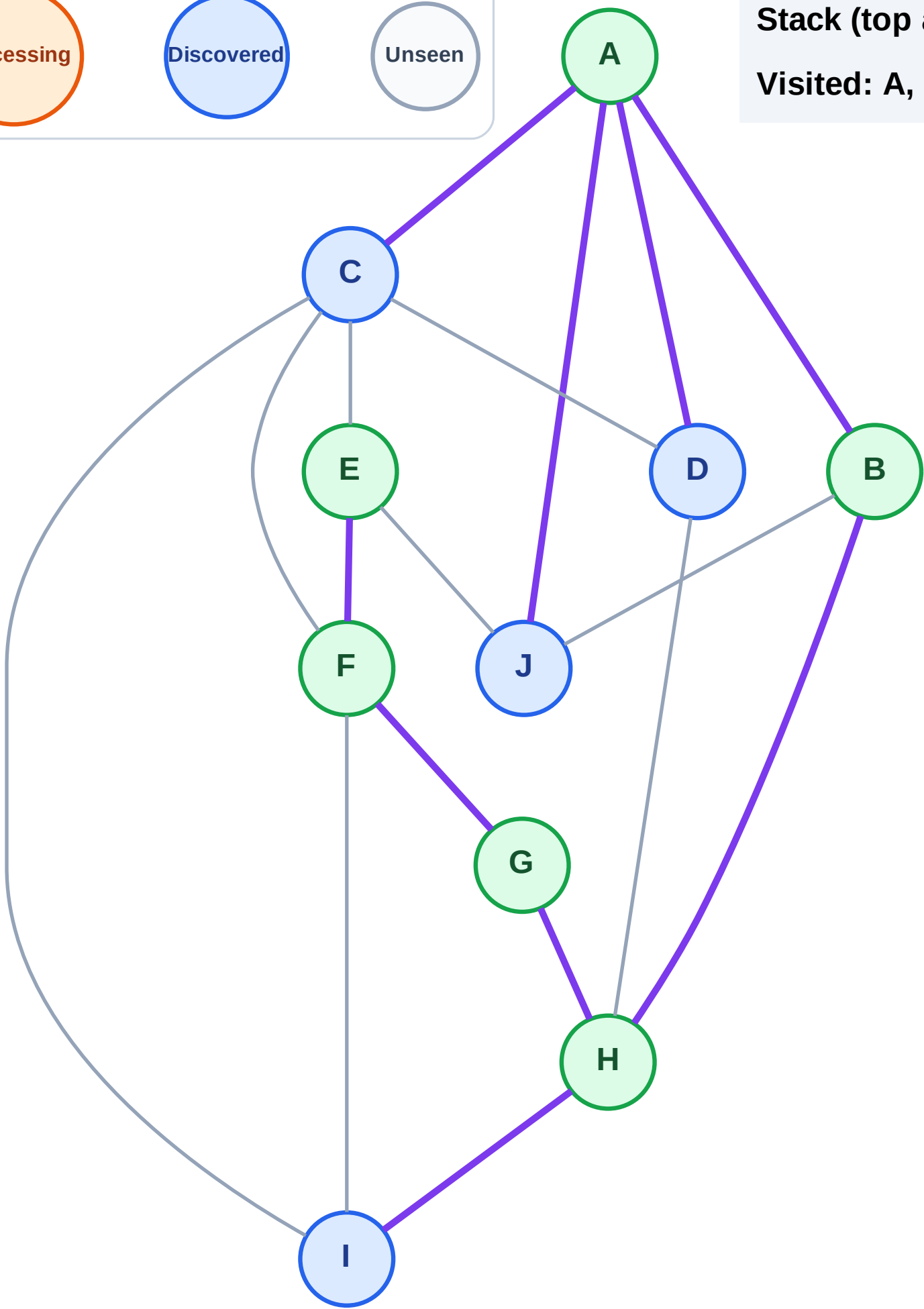
Step 13: No new neighbors from E

Legend



Stack (top at right): J | D | C | I

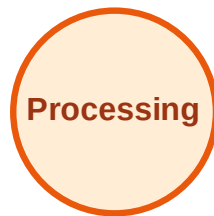
Visited: A, B, E, F, G, H



DFS Traversal

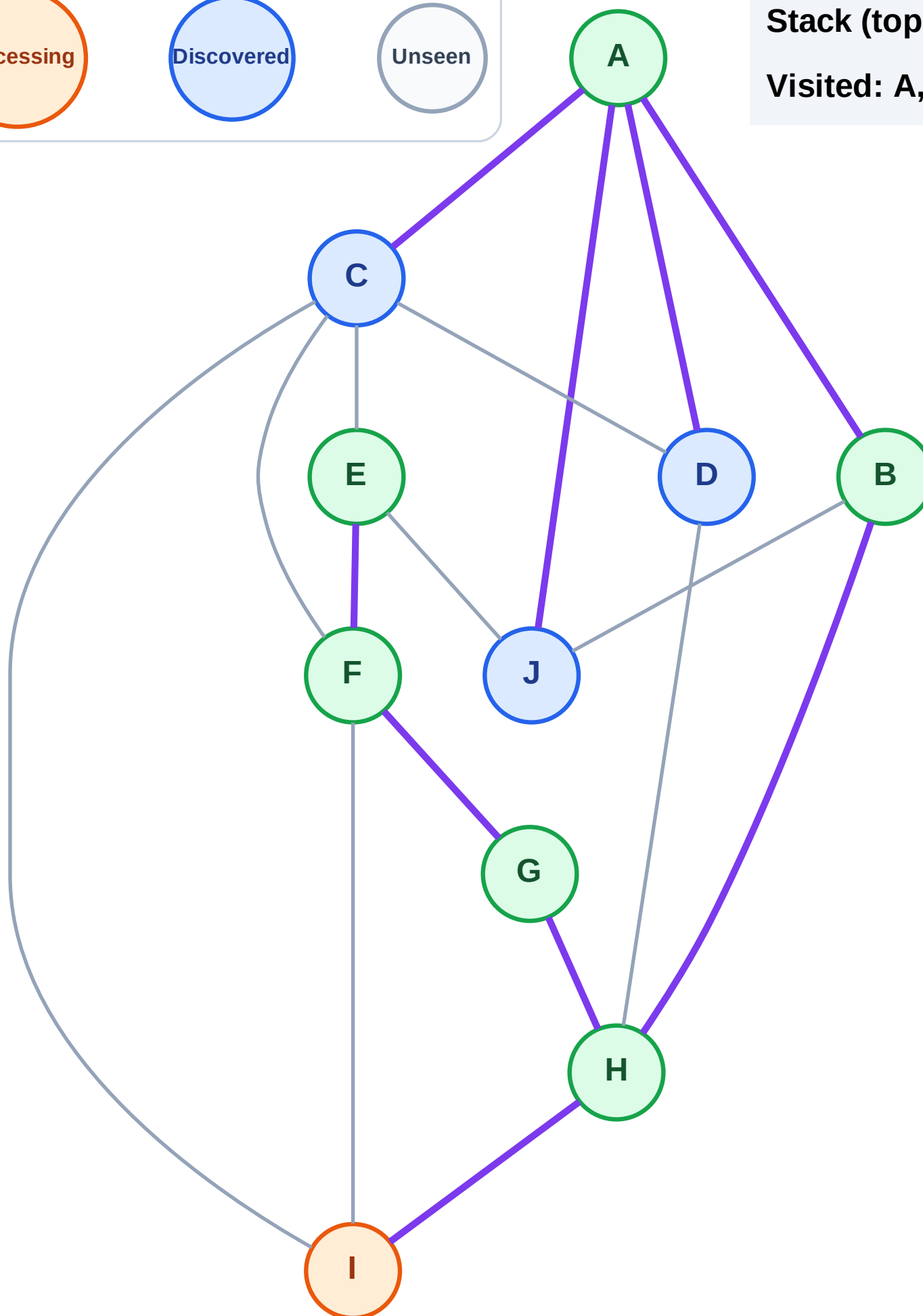
Step 14: Process node I

Legend



Stack (top at right): J | D | C

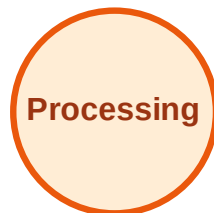
Visited: A, B, E, F, G, H



DFS Traversal

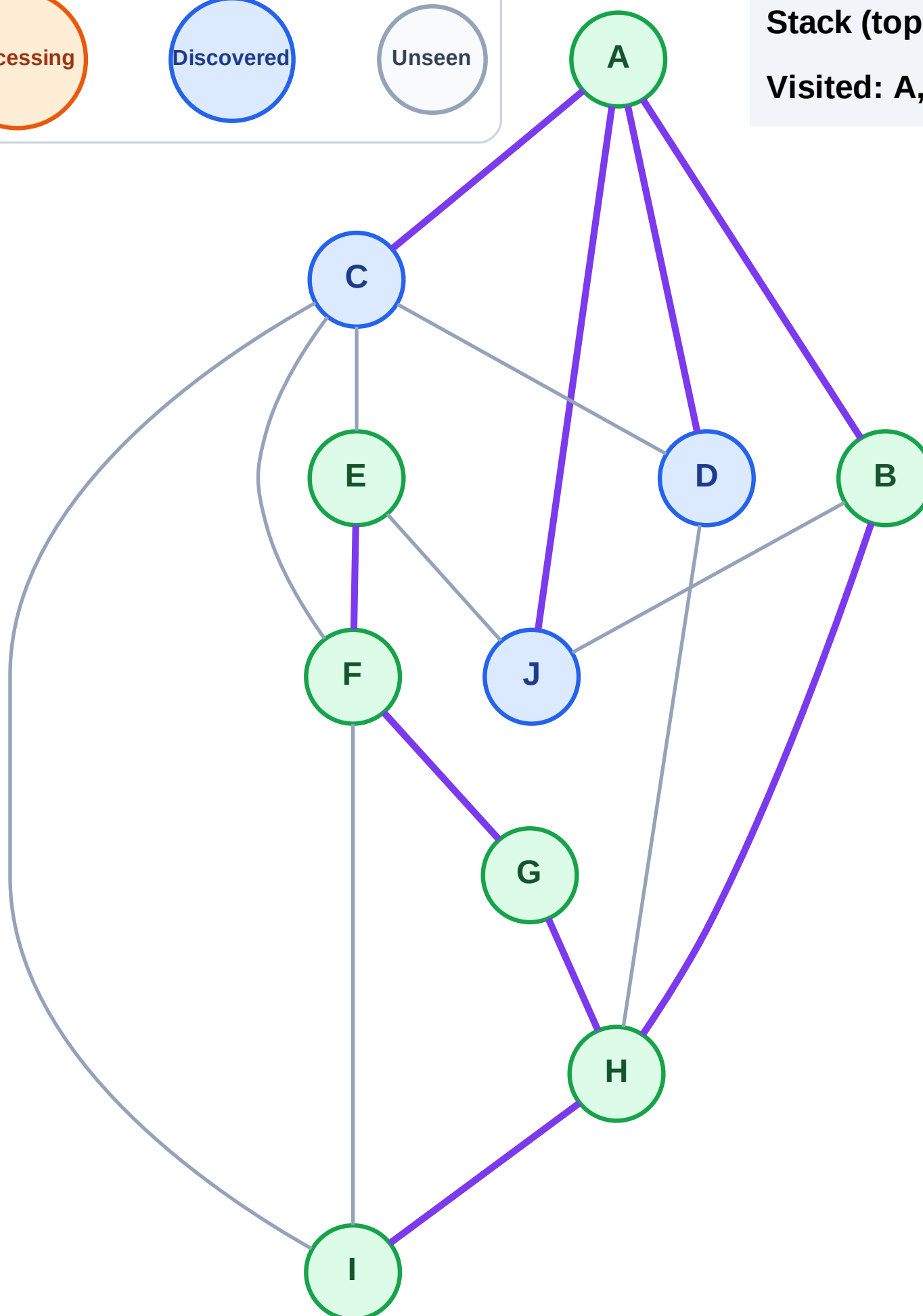
Step 15: No new neighbors from I

Legend



Stack (top at right): J | D | C

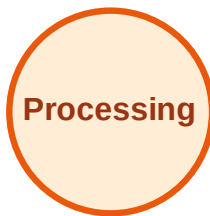
Visited: A, B, E, F, G, H, I



DFS Traversal

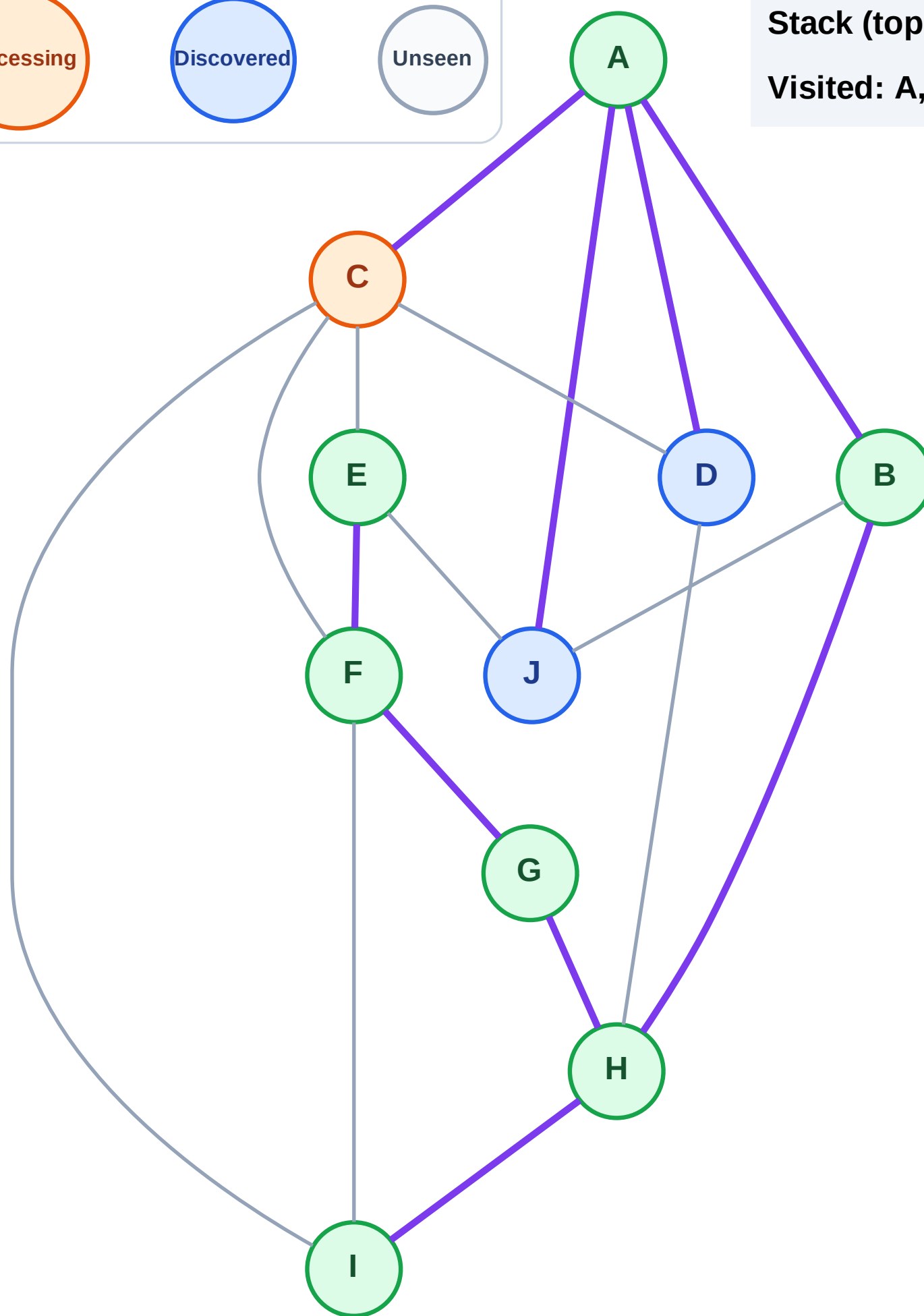
Step 16: Process node C

Legend



Stack (top at right): J | D

Visited: A, B, E, F, G, H, I



DFS Traversal

Step 17: No new neighbors from C

Legend

Complete

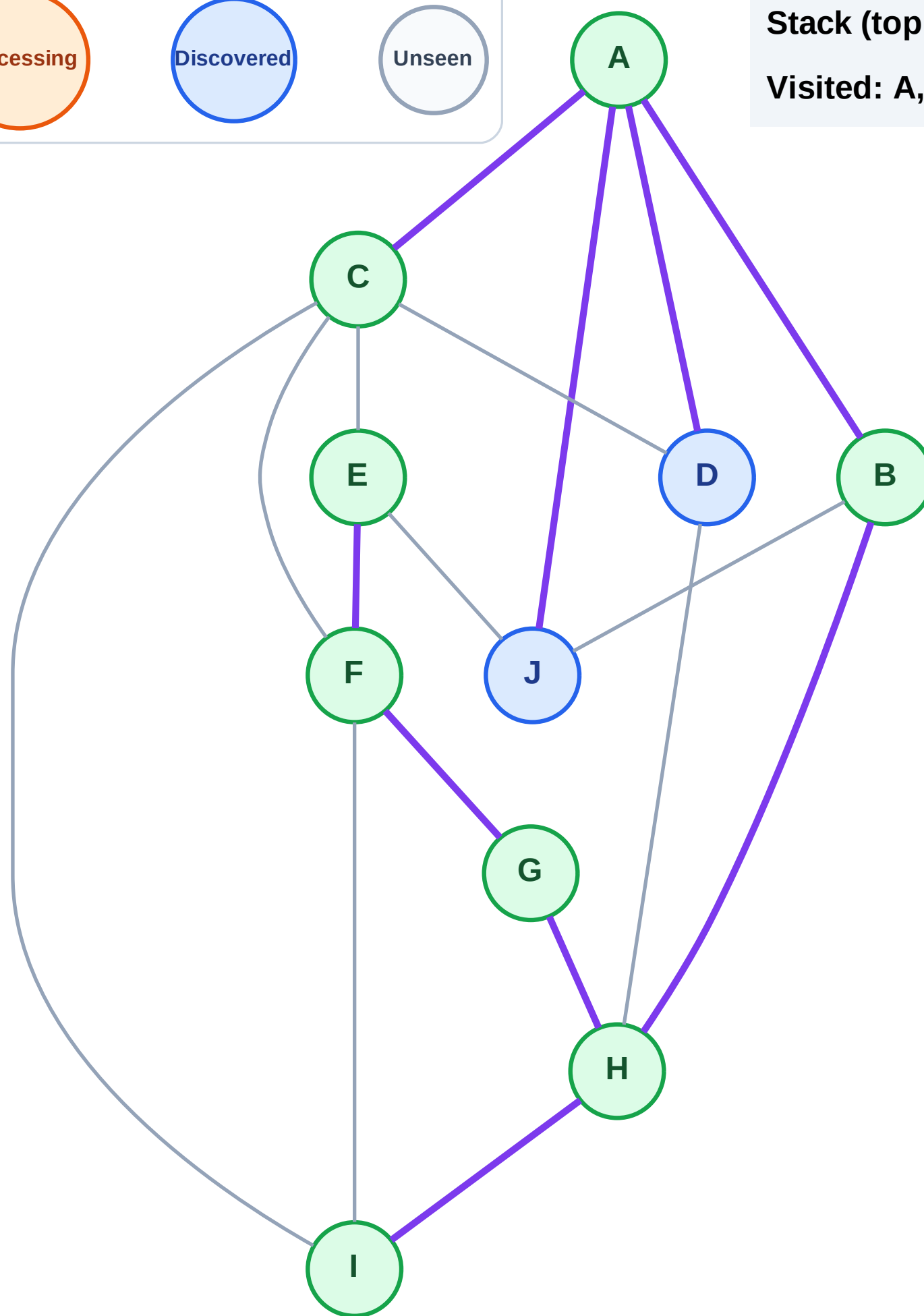
Processing

Discovered

Unseen

Stack (top at right): J | D

Visited: A, B, C, E, F, G, H, I



DFS Traversal

Step 18: Process node D

Legend

Complete

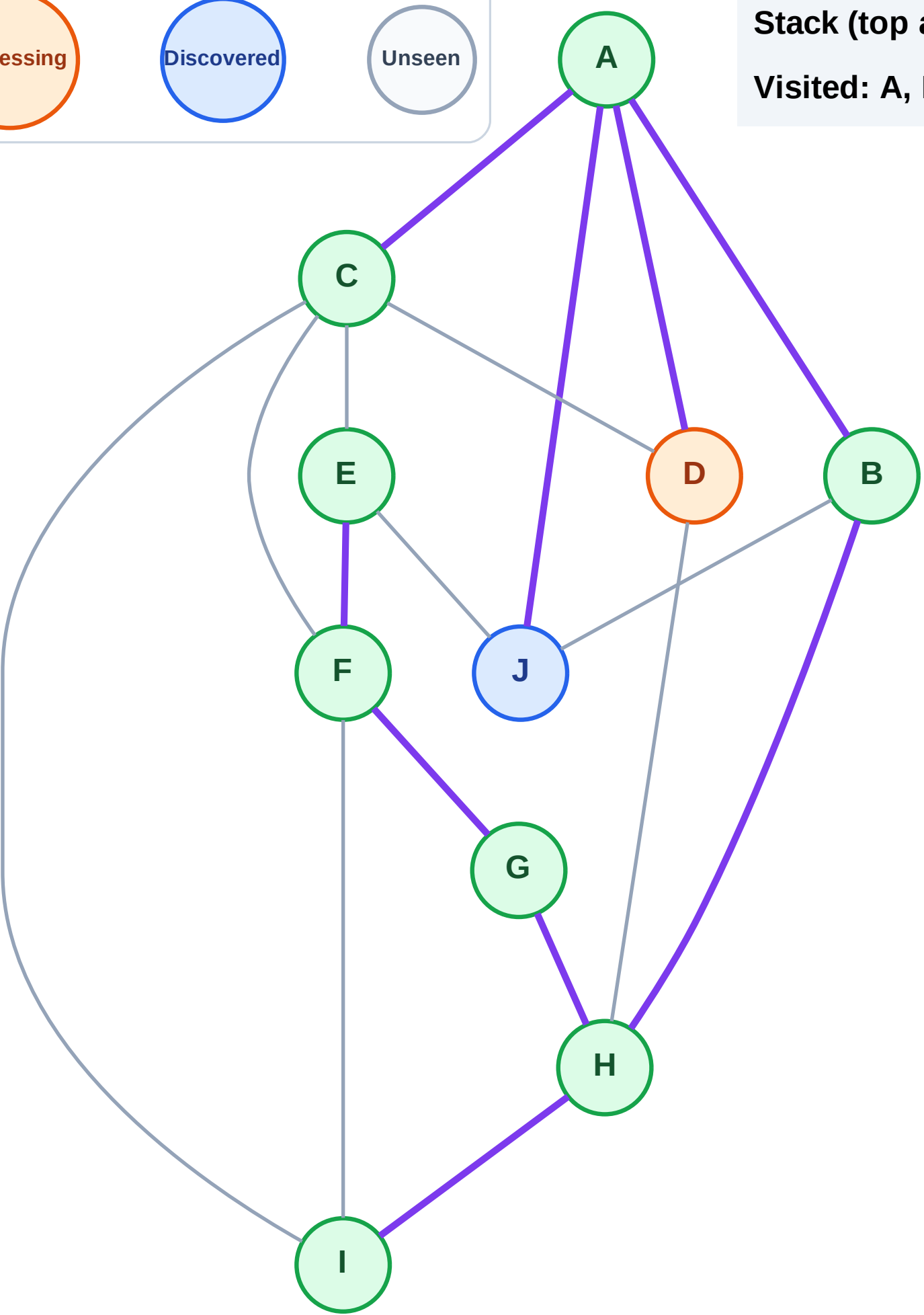
Processing

Discovered

Unseen

Stack (top at right): J

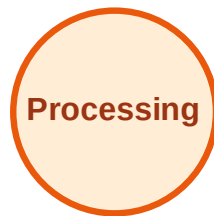
Visited: A, B, C, E, F, G, H, I



DFS Traversal

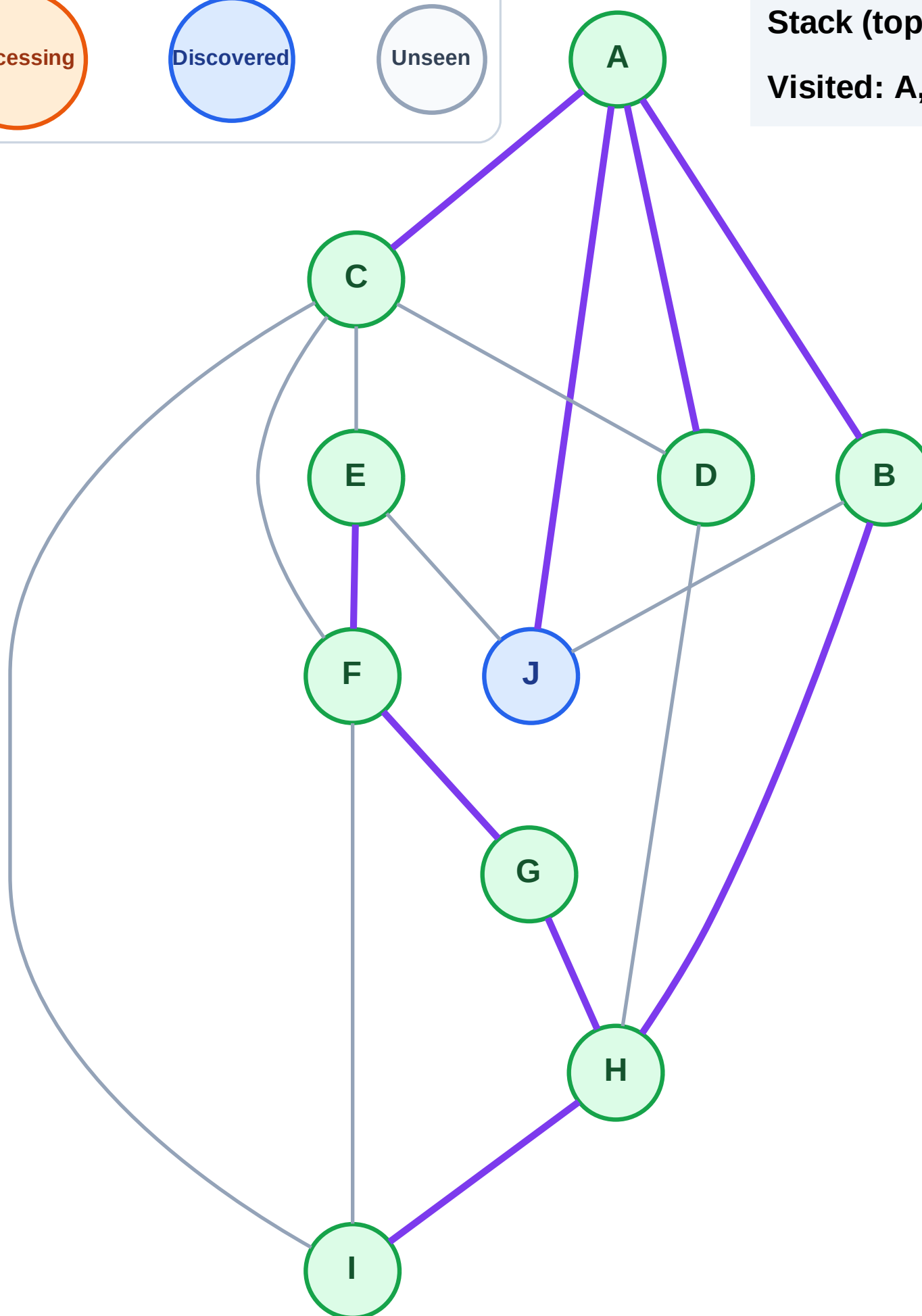
Step 19: No new neighbors from D

Legend



Stack (top at right): J

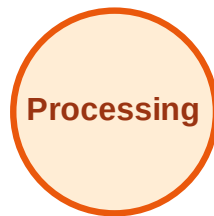
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

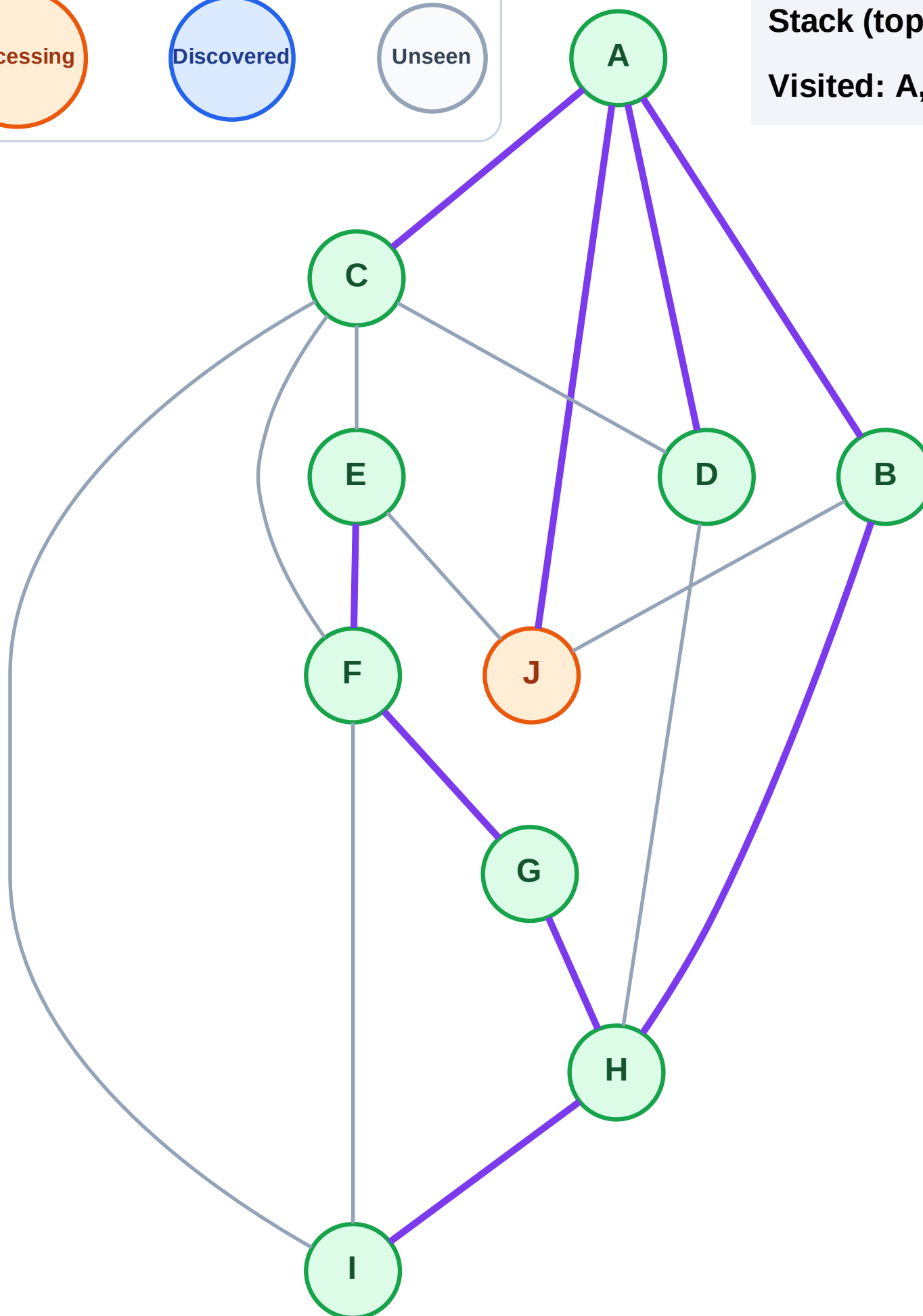
Step 20: Process node J

Legend



Stack (top at right): (empty)

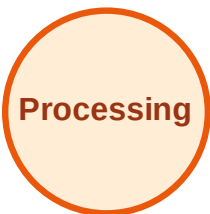
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

